

SQIsign2DPush: Faster Signature Scheme Using 2-Dimensional Isogenies

Kohei Nakagawa¹ and Hiroshi Onuki²

¹ NTT Social Informatics Laboratories, Japan

² The University of Tokyo, Japan

Abstract. Isogeny-based cryptography involves cryptographic schemes whose security is based on the hardness of a mathematical problem called the isogeny problem and is attracting attention as one of the candidates for post-quantum cryptography. A representative isogeny-based cryptography is the signature scheme called SQIsign, which was submitted to the NIST PQC standardization competition for additional signature. SQIsign has attracted much attention because of its very short signature and key size among candidates for the NIST PQC standardization. Recently, many new signature schemes using high-dimensional isogenies have been proposed, such as SQIsignHD, SQIsign2D-West, SQIsign2D-East, and SQIPrime. Last year, SQIsign advanced to Round 2 of the NIST competition and was updated to version 2.0 (we call it SQIsign-v2.0), which is based on SQIsign2D-West. SQIsign-v2.0 achieves smaller signature sizes and faster verification. However, the signing costs are relatively high. In this paper, we propose a new signature scheme named ‘SQIsign2DPush’, which has a smaller signing cost than SQIsign-v2.0 while the signature size and the verification cost are almost the same.

1 Introduction

In recent years, isogeny-based cryptography has been actively studied as one of the candidates for post-quantum cryptography (PQC). One of the representative isogeny-based cryptosystems is the signature scheme called SQIsign [15], which was submitted to the NIST PQC standardization competition for additional signature [10]. SQIsign has attracted much attention because of its very short signature and key size among the candidates for the NIST PQC additional signature.

SIDH [26] and SIKE [25] were also representative isogeny-based cryptosystems. However, recent attacks [9,30,34] broke the security of SIDH and SIKE. These attacks show that any isogeny can be efficiently recovered from its evaluation on a sufficiently large torsion subgroup by computing high-dimensional isogenies. In response, a lot of cryptographic applications using high-dimensional isogenies have been studied: encryption schemes such as FESTA [6], QFESTA [31], and POKÉ [5], and variants of SQIsign such as SQIsignHD [13], SQIsign2D-West [4], SQIsign2D-East [32], and SQIPrime [19].

SQIsignHD is the first signature scheme using high-dimensional isogenies and achieves much smaller signature sizes and higher signing performance than SQIsign. However, it requires 4 or 8-dimensional isogeny computations for verification, which leads to a large verification cost. On the other hand, SQIsign2D-West, SQIsign2D-East, and SQIPrime use 2-dimensional isogenies for verification. As a result, all of these schemes achieved high verification speeds, although the signing costs have slowed down compared to SQIsignHD. In particular, SQIsign2D-West was adopted as the updated version of SQIsign [1] (we call it SQIsign-v2.0) when SQIsign advanced to Round 2 of the NIST competition. However, the signing cost of SQIsign-v2.0 remains relatively slow compared to other NIST candidates. One of the dominant factors in the signing cost of SQIsign-v2.0 is the computation of an isogeny called the ‘auxiliary isogeny’. This is an isogeny of a certain (generally non-smooth) degree from a given elliptic curve. In SQIsign-v2.0, many 2-dimensional isogenies are required to compute this auxiliary isogeny.

Recent Works. We introduce two recent works on isogeny-based signatures that were published after the initial version of our paper. One is a paper that proposed a new signature scheme called SQIsign2D² [39], which is based on SQIsign2D-East and improves its computational efficiency. The other is a paper about an algorithm called Qlapoti [8]. By using Qlapoti as a building block, the signing cost of SQIsign-v2.0 is significantly reduced. Though the initial objective of this paper was to establish a more efficient signature scheme than the original SQIsign-v2.0, we also take into account SQIsign2D² and SQIsign-v2.0 with Qlapoti in the efficiency comparisons in Section 6.

1.1 Contributions

In this paper, we make the following contributions.

- In Section 3, we construct a new algorithm named **PushRandIsog**, which computes an auxiliary isogeny, i.e., a fixed (generally non-smooth) degree isogeny from a given elliptic curve E .
- Using **PushRandIsog** as a building block, we construct a new variant of SQIsign in Section 4. We name our new signature scheme ‘SQIsign2DPush’.
- In Section 5, we propose a new theorem (Theorem 1), which can be used to prove the EUF-CMA security of signature under the *Fiat-Shamir with abort* framework [29] and the *hint-assisted wHVZK* framework [1]. This theorem can be utilized for the security proof that accounts for the failure scenarios of SQIsign-v2.0, as stated in [2, Section 7].
- Using the above theorem, we prove the EUF-CMA security of SQIsign2D-Push (Theorem 4).
- In Section 6, we give concrete parameters of SQIsign2DPush for the NIST security level 1, 3, and 5. Under these parameter settings, we analyze the signature sizes and computational costs, comparing them with several SQIsign variants, including SQIsign-v2.0.

- Our SQIsign2DPush has the lowest signing cost among the SQIsign variants before Qlapoti while the signature sizes and the verification costs are comparable.

Our SQIsign2DPush is based mainly on SQIsignHD [13] rather than SQIsign-v2.0 [1]. In fact, the key generation, the commitment, and the challenge are almost the same as SQIsignHD. In the response, however, we compute an *auxiliary isogeny* of a fixed (probably non-smooth) degree from a given curve as in SQIsign-v2.0. To compute such an isogeny, we construct a new algorithm named **PushRandIsog**. Let E_0 be a supersingular elliptic curve whose endomorphism ring is special p -extremal [28, Section 2.3]. Our **PushRandIsog** takes an integer d and an isogeny $\psi : E_0 \rightarrow E$ as input, and outputs a d -isogeny φ from E . Our main idea for **PushRandIsog** is to compute a d -isogeny φ_0 from E_0 using the idea of QFESTA [31, Algorithm 2] and then *push* the isogeny φ_0 by the given isogeny $\psi : E_0 \rightarrow E$. Then we obtain a d -isogeny φ from E . For this, we use **FastCommit** from SQIsignHD [13, Algorithm 1] and a technique from POKÉ [5].

2 Preliminaries

In this section, we summarize some background knowledge used in this paper. Definitions of fundamental cryptographic concepts, such as the Σ -protocol and signature scheme, are provided in Appendix A.

2.1 Notation

Throughout this paper, we use the following notation. We let p be a prime number of cryptographic size, i.e. p is at least about 2^{256} and let λ be a security parameter. For a finite set S , we write $x \xleftarrow{\$} S$ if x is sampled uniformly at random from S . Let $\perp$ be the symbol that indicates the failure of an algorithm. We denote the neutral element of an elliptic curve by O .

2.2 Abelian varieties and Isogenies

In this paper, we mainly use principally polarized superspecial abelian varieties defined over a finite field of characteristic p of dimension one or two. Such a variety is isomorphic to a supersingular elliptic curve, the product of two supersingular elliptic curves, or a Jacobian of a superspecial hyperelliptic curve of genus two, and always has a model defined over $\mathbb{F}_{p^2}$. Therefore, we only consider varieties defined over $\mathbb{F}_{p^2}$.

Basic Facts. An *isogeny* is a rational map between abelian varieties which is a surjective group homomorphism with finite kernel. The *degree* of an isogeny φ is its degree as a rational map and is denoted by $\deg \varphi$. An isogeny φ is *separable* if $\# \ker \varphi = \deg \varphi$. A separable isogeny is uniquely determined by its kernel up to

post-composition of isomorphism. For an isogeny $\varphi : A \rightarrow B$ between principally polarized abelian varieties, there exists an isogeny $B \rightarrow A$ called the *dual isogeny* (with respect to the polarizations). We denote the dual isogeny of φ by $\hat{\varphi}$.

Let $\varphi : A \rightarrow B$ and $\psi : A \rightarrow C$ be separable isogenies of coprime degrees. If a separable isogeny $\psi' : B \rightarrow D$ satisfies $\ker \psi' = \varphi(\ker \psi)$, we say that ψ' is the *push-forward* of ψ by φ and denote it by $\psi' = [\varphi]_* \psi$. Under the same situation, we say that ψ is the *pull-back* of ψ' by φ and denote it by $\psi = [\varphi]^* \psi'$.

Let A and B be principally polarized abelian varieties. If there exists an isogeny between A and B , then the dimensions of A and B are the same. If A is superspecial, then there exists an isogeny between A and B if and only if B is a superspecial abelian variety of the same dimension as A .

Let A be a principally polarized abelian variety and ℓ a positive integer. An ℓ -isotropic subgroup of A is a subgroup of the ℓ -torsion subgroup $A[\ell]$ of A on which the ℓ -Weil pairing is trivial. An ℓ -isotropic subgroup G is *maximal* if there is no other ℓ -isotropic subgroup containing G . A separable isogeny whose kernel is a maximal ℓ -isotropic subgroup is called an ℓ -isogeny if the dimension of the domain is one or an (ℓ, ℓ) -isogeny if the dimension of the domain is two.

Let E be an elliptic curve defined over $\mathbb{F}_{p^2}$ and let n be an integer such that the n -torsion subgroup $E[n]$ is contained in $E(\mathbb{F}_{p^2})$. Then we can compute a deterministic basis of $E[n]$ by using deterministic algorithms such as [10, Algorithm 3] and [1, Algorithm 2.2].

Computing Isogenies. Let A be a principally polarized abelian variety, ℓ a positive integer, and G a maximal ℓ -isotropic subgroup of A .

If the dimension of A is one, then we can compute an ℓ -isogeny φ with kernel G by Vélú's formulas [37]. More precisely, given A , ℓ , G , Vélú's formulas give a method to compute the codomain of φ in $O(\ell)$ operations on a field containing the points in G . In addition, for additional input $P \in A$, we can compute $\varphi(P)$ in $O(\ell)$ operations on a field containing the points in G and P .

If the dimension of A is two and $\ell = 2$, then we can compute $(2, 2)$ -isogeny with kernel G by the recent formulas proposed by Dartois, Maino, Pope, and Robert [14].

For an isogeny $\varphi : A \rightarrow B$, we say that information $\mathcal{I}_\varphi$ is an *efficient representation* of φ when we can compute $\varphi(P)$ in polynomial time from a given point $P \in A$ and the information $\mathcal{I}_\varphi$. For example, the tuple (A, ℓ, G) described above is an efficient representation of ℓ -isogeny $\varphi : A \rightarrow B$ when ℓ is smooth. In this paper, we assume that an isogeny is given by an efficient representation.

2.3 Quaternion Algebras and the Deuring Correspondence

Quaternion Algebras. A *quaternion algebra* over $\mathbb{Q}$ is a division algebra defined by $\mathbb{Q} + \mathbb{Q}\mathbf{i} + \mathbb{Q}\mathbf{j} + \mathbb{Q}\mathbf{k}$ and $\mathbf{i}^2 = a, \mathbf{j}^2 = b, \mathbf{ij} = -\mathbf{ji} = \mathbf{k}$ for $a, b \in \mathbb{Q}^*$. We denote it by $H(a, b)$. We say $H(a, b)$ is *ramified* at a place v of $\mathbb{Q}$ if $H(a, b) \otimes_{\mathbb{Q}} \mathbb{Q}_v$ is not isomorphic to the algebra of the 2×2 matrices over $\mathbb{Q}_v$. There exists a

quaternion algebra ramified exactly at p and ∞ . Such an algebra is unique up to isomorphism. We denote it by $\mathcal{B}_{p,\infty}$. In this paper, we only consider $\mathcal{B}_{p,\infty}$.

Let $\alpha = x + y\mathbf{i} + z\mathbf{j} + t\mathbf{k} \in \mathcal{B}_{p,\infty}$ with $x, y, z, t \in \mathbb{Q}$. The *conjugate* of α is $x - y\mathbf{i} - z\mathbf{j} - t\mathbf{k}$ and is denoted by $\bar{\alpha}$. The *reduced norm* of α is $\alpha\bar{\alpha}$ and is denoted by $n(\alpha)$. In this paper, we simply call it the *norm*.

An *order* $\mathcal{O}$ of $\mathcal{B}_{p,\infty}$ is a subring of $\mathcal{B}_{p,\infty}$ that is also a $\mathbb{Z}$ -lattice of rank 4. This means that $\mathcal{O} = \mathbb{Z}\alpha_1 + \mathbb{Z}\alpha_2 + \mathbb{Z}\alpha_3 + \mathbb{Z}\alpha_4$ for a basis $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ of $\mathcal{B}_{p,\infty}$. We denote such an order by $\mathbb{Z}\langle\alpha_1, \alpha_2, \alpha_3, \alpha_4\rangle$. An order $\mathcal{O}$ is said to be *maximal* if there is no larger order that contains $\mathcal{O}$.

For a maximal order $\mathcal{O}$, a (integral) *left $\mathcal{O}$ -ideal* I is a $\mathbb{Z}$ -lattice of rank 4 satisfying $I \subseteq \mathcal{O}$ and $\mathcal{O} \cdot I \subseteq I$. A *right $\mathcal{O}$ -ideal* is similarly defined. For an ideal I , we denote its conjugate by $\bar{I} = \{\bar{\alpha} \mid \alpha \in I\}$. We denote by $n(I)$ the *reduced norm* of ideal I , defined as (the unique positive generator of) the $\mathbb{Z}$ -module generated by the norms of the elements of I . In this paper, we simply call it the *norm*.

The *ideal equivalence* denoted by $I \sim J$ means that there exists $\beta \in \mathcal{B}_{p,\infty}^*$ such that $J = I\beta$. In particular, the ideal $J = I \frac{\bar{\alpha}}{n(I)}$ for $\alpha \in I$ is equivalent to I and has norm $n(\alpha)/n(I)$. In this paper, we call the value $n(\alpha)/n(I)$ the *normalized norm* of $\alpha \in I$ and denote it by $q_I(\alpha)$. It is known that there always exists $\alpha \in I$ such that $q_I(\alpha) \leq 2\sqrt{2p}/\pi$, that is, there exists an ideal $J \sim I$ of norm $\leq 2\sqrt{2p}/\pi$ [13, Lemma 12].

Deuring Correspondence. Deuring [18] showed that the endomorphism ring of a supersingular elliptic curve over $\mathbb{F}_{p^2}$ is isomorphic to a maximal order of $\mathcal{B}_{p,\infty}$ and gave a correspondence (*Deuring correspondence*) where a supersingular elliptic curve E over $\mathbb{F}_{p^2}$ corresponds to a maximal order isomorphic to $\text{End}(E)$.

Suppose $p \equiv 3 \pmod{4}$. This is the setting we use in our protocol. Then we can take $\mathcal{B}_{p,\infty} = H(-1, -p)$ and an elliptic curve over $\mathbb{F}_{p^2}$ with j -invariant 1728 is supersingular. Let E_0 be the elliptic curve over $\mathbb{F}_{p^2}$ defined by $y^2 = x^3 + x$. Then $j(E_0) = 1728$, so E_0 is supersingular. We define the endomorphisms $\iota : (x, y) \mapsto (-x, \sqrt{-1}y)$ and $\pi : (x, y) \mapsto (x^p, y^p)$ of E_0 , where $\sqrt{-1}$ is a fixed square root of -1 in $\mathbb{F}_{p^2}$. The endomorphism ring of E_0 is isomorphic to $\mathcal{O}_0 := \mathbb{Z}\langle 1, \mathbf{i}, \frac{1+\mathbf{j}}{2}, \frac{1+\mathbf{k}}{2} \rangle$. An isomorphism is given by $\iota \mapsto \mathbf{i}$ and $\pi \mapsto \mathbf{j}$. From now on, we assume that $\text{End}(E_0)$ and $\mathcal{O}_0$ always correspond to each other through this isomorphism.

Let $P \in E_0(\mathbb{F}_{p^2})$ and $\alpha = x + y\mathbf{i} + z\frac{1+\mathbf{j}}{2} + t\frac{1+\mathbf{k}}{2} \in \mathcal{O}_0$ for $x, y, z, t \in \mathbb{Z}$. Given P and x, y, z, t , we can compute $\alpha(P)$ in $\tilde{O}(\log \max\{|x|, |y|, |z|, |t|\})$ operations on $\mathbb{F}_{p^2}$ and $O(\log p)$ operations on $\mathbb{F}_{p^4}$. The latter operations on $\mathbb{F}_{p^4}$ are necessary only for the case when the order of P is even. We need to compute $\alpha(P_0)$ and $\alpha(Q_0)$ for a fixed torsion basis P_0, Q_0 of E_0 in our protocol. In this case, by pre-computing the images of P_0 and Q_0 under $\mathbf{i}, \frac{1+\mathbf{j}}{2}$, and $\frac{1+\mathbf{k}}{2}$, we can compute $\alpha(P_0)$ and $\alpha(Q_0)$ by scalar multiplications by x, y, z, t and additions.

The Deuring correspondence also gives a correspondence between isogenies and ideals. Let E_1 be a supersingular elliptic curve over $\mathbb{F}_{p^2}$ and let $\mathcal{O}_1$ be a maximal order of $\mathcal{B}_{p,\infty}$ such that $\mathcal{O}_1 \cong \text{End}(E_1)$. We fix an isomorphism ι between $\text{End}(E_1)$ and $\mathcal{O}_1$. Let $\varphi : E_1 \rightarrow E_2$ be a d -isogeny, then the isogeny

φ can be associated to a left $\mathcal{O}_1$ -ideal I_φ of norm d via the isomorphism ι as follows:

$$I_\varphi = \{\iota(\psi \circ \varphi) \mid \psi \in \text{Hom}(E_2, E_1)\}.$$

This ideal I_φ is also a right $\mathcal{O}_2$ -ideal for a maximal order $\mathcal{O}_2$ satisfying $\mathcal{O}_2 \cong \text{End}(E_2)$. In particular, φ induces an isomorphism from $\text{End}(E_2)$ to $\mathcal{O}_2$ defined by

$$\alpha \mapsto \frac{1}{\deg \varphi} \iota(\hat{\varphi} \circ \alpha \circ \varphi).$$

Moreover, two isogenies $\varphi, \psi : E_1 \rightarrow E_2$ with the same domain and codomain correspond to the equivalent ideals $I_\varphi \sim I_\psi$. On the contrary, a left $\mathcal{O}_1$ -ideal I can be associated to an isogeny φ_I from E_1 with the following kernel:

$$\ker \varphi_I = \bigcap_{\alpha \in I} \ker(\iota^{-1}(\alpha)).$$

2.4 Kani's Lemma

In this subsection, we introduce Kani's lemma [27], which is used in the attacks to SIDH [9,30,34].

Theorem 1 ([30, Theorem 1]) *Let d_1, d_2 , and D be pairwise coprime integers such that $D = d_1 + d_2$, and let E, E_1, E_2 , and F be elliptic curves connected by the following diagram of isogenies:*

$$\begin{array}{ccc} E & \xrightarrow{\varphi_2} & E_2 \\ \varphi_1 \downarrow & \nearrow f & \downarrow \varphi'_1 \\ E_1 & \xrightarrow{\varphi'_2} & F \end{array}$$

where $\varphi'_2 \circ \varphi_1 = \varphi'_1 \circ \varphi_2$, $f = \varphi_2 \circ \hat{\varphi}_1$, and $\deg(\varphi_i) = \deg(\varphi'_i) = d_i$ for $i \in \{1, 2\}$. Then the map

$$\Phi = \begin{pmatrix} \hat{\varphi}_1 & -\hat{\varphi}_2 \\ \varphi'_2 & \varphi'_1 \end{pmatrix} : E_1 \times E_2 \rightarrow E \times F \quad (1)$$

is a (D, D) -isogeny with respect to the natural product polarizations on $E_1 \times E_2$ and $E \times F$, and has kernel $\{([d_2]P, f(P)) \mid P \in E_1[D]\}$. Conversely, a (D, D) -isogeny with this kernel is equal to Φ up to isomorphism.

2.5 Existing Algorithms

In this subsection, we introduce the existing algorithms used in this paper. Let p be a prime such that $p \equiv 3 \pmod{4}$ and let $E_0/\mathbb{F}_{p^2}$ be a supersingular elliptic curve defined by $y^2 = x^3 + x$. We also write $\mathcal{O}_0 = \mathbb{Z}\langle 1, \mathbf{i}, \frac{1+\mathbf{j}}{2}, \frac{1+\mathbf{k}}{2} \rangle$, which is a maximal order of $\mathcal{B}_{p,\infty}$ isomorphic to $\text{End}(E_0)$.

FullRepresentInteger $_{\mathcal{O}_0}(M)$ [16, Algorithm 1]: This algorithm takes an integer $M > p$ as input and outputs $\alpha \in \mathcal{O}_0$ of norm M . From the Deuring correspondence, we can see this α as an endomorphism of E_0 of degree M .

FastDoublePath $_{2^{e_1}, 3^{e_2}}$ /**FastCommit** $_{2^{e_1}, 3^{e_2}}$ [13, Algorithm 1]: These algorithms are used in the setting where $p = c \cdot 2^{e_1} 3^{e_2} - 1$. Then **FastDoublePath** takes 2^{e_1} and 3^{e_2} as input and outputs a 2^{2e_1} -isogeny $\varphi_1 : E_0 \rightarrow E$, a 3^{2e_2} -isogeny $\varphi_2 : E_0 \rightarrow E$, and their corresponding left $\mathcal{O}_0$ -ideals I_1, I_2 , where these two isogenies have the same codomain E . **FastCommit** is similar to **FastDoublePath**, but it only outputs a 3^{2e_2} -isogeny $\varphi_2 : E_0 \rightarrow E$ and its corresponding ideal I_2 . We decompose φ_1 into $\varphi_1'' \circ \varphi_1'$ such that $\deg \varphi_1' = \deg \varphi_1'' = 2^{e_1}$. Then φ_1 is efficiently represented by generators of $\ker \varphi_1'$ and $\ker \varphi_1''$ since $E_0[2^{e_1}]$ is $\mathbb{F}_{p^2}$ -rational. φ_2 is represented similarly.

IsogToIdeal $(\psi, \varphi, I_\varphi)$ [13, Algorithm 10]: This is an algorithm to convert an isogeny to its corresponding ideal. Such an algorithm is used broadly in many isogeny-based cryptosystems. In this paper, we use the algorithm proposed in SQIsignHD. It takes as input an isogeny ψ , an isogeny φ from E_0 to the domain of ψ , and a left $\mathcal{O}_0$ -ideal I_φ corresponding to φ , where $\deg \psi$ is smooth and coprime to $\deg \varphi$ and $E_0[\deg \psi]$ is in $E_0(\mathbb{F}_{p^2})$. Then it outputs the ideal I_ψ corresponding to ψ via the isomorphism induced by φ .

KaniEval $(d_1, d_2, P_1, Q_1, P_2, Q_2; \mathcal{S}_1; \mathcal{S}_2)$ [32, Section 2.4]: Here, we use the same notation as in Theorem 1. According to Theorem 1, we can evaluate a d_1 -isogeny $\hat{\varphi}_1$ and a d_2 -isogeny $\hat{\varphi}_2$ from the knowledge of $\varphi_2 \circ \hat{\varphi}_1|_{E_1[D]}$. We denote the algorithm by **KaniEval**. More precisely, **KaniEval** takes coprime integers d_1, d_2 , a basis P_1, Q_1 of $E_1[D]$, $P_2 = \varphi_2 \circ \hat{\varphi}_1(P_1), Q_2 = \varphi_2 \circ \hat{\varphi}_1(Q_1) \in E_2[D]$, a finite subset $\mathcal{S}_1 \subseteq E_1$, and a finite subset of $\mathcal{S}_2 \subseteq E_2$ as input. Then **KaniEval** returns the codomain E of $\hat{\varphi}_1$ and $\hat{\varphi}_2$, the image of $\mathcal{S}_1$ under $\hat{\varphi}_1$, and the image of $\mathcal{S}_2$ under $\hat{\varphi}_2$. In this algorithm, we compute a (D, D) -isogeny, which can be computed efficiently when D is smooth. In this paper, we consider the case where D is a power of 2.

2.6 Signature Schemes in High-Dimension

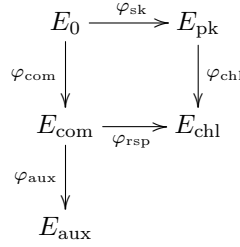
Here, we briefly describe some signature schemes using high-dimensional isogenies: SQIsignHD, SQIsign-v2.0 (originally SQIsign2D-West), SQIsign2D-East, and SQIsign2D².

SQIsignHD is the first signature scheme using high-dimensional isogenies. It has short signature size and fast signing procedure but its verification requires 4 or 8-dimensional isogeny computations. SQIsign2D-West and SQIsign2D-East are signature schemes proposed simultaneously, both of which use only 1 and 2-dimensional isogeny computations for verification. As a result, all of these schemes achieved high verification speeds, although the signing costs have slowed down compared to SQIsignHD. In particular, SQIsign2D-West was adopted as

Table 1. The algorithms each protocol uses to compute each isogeny.

Protocol	φ_{sk}	φ_{com}	φ_{aux}
SQIsignHD	FastDoublePath	FastCommit	-
SQIsign-v2.0	IdealToIsogeny [4, Algorithm 2]	IdealToIsogeny	IdealToIsogeny
SQIsign2D-East	RandIsogImg [31, Algorithm 2]	RandIsogImg	GenRandIsogImg [32, Algorithm 2]
SQIsign2D ²	ImRanIso [39, Algorithm 1]	ImRanIso	GenImRanIso [39, Algorithm 2]

the updated version of SQIsign [1] (we call it SQIsign-v2.0) when SQIsign advanced to Round 2 of the NIST competition. Finally, SQIsign2D² is a speed-up version of SQIsign2D-East. All of these schemes are roughly based on the Σ -protocol using the following diagram.



The procedure of the Σ -protocol is roughly as follows. A prover having a secret $\varphi_{\text{sk}} : E_0 \rightarrow E_{\text{pk}}$ first generates a random isogeny $\varphi_{\text{com}} : E_0 \rightarrow E_{\text{com}}$ and sends E_{com} to a verifier. The verifier then generates a random isogeny $\varphi_{\text{chl}} : E_{\text{pk}} \rightarrow E_{\text{chl}}$ and returns it to the prover. The prover then computes an isogeny $\varphi_{\text{rsp}} : E_{\text{com}} \rightarrow E_{\text{chl}}$ and sends φ_{rsp} to the verifier. At this time, the prover also sends an auxiliary isogeny φ_{aux} to help the verifier computations (except for SQIsignHD). Finally, the verifier checks if φ_{rsp} is an isogeny from E_{com} to E_{chl} .

The main differences among SQIsignHD, SQIsign-v2.0, SQIsign2D-East, and SQIsign2D² lie in the algorithms they use to generate φ_{sk} , φ_{com} , and φ_{aux} . In addition, in SQIsignHD, the prover does not compute the auxiliary isogeny φ_{aux} ; instead, the prover sends only the efficient representation of φ_{rsp} to the verifier. As a result, the verifier requires 4- or 8-dimensional isogeny computation to reconstruct φ_{rsp} from its efficient representation. We summarize in Table 1 the algorithms used in each protocol to compute φ_{sk} , φ_{com} , and φ_{aux} .

The different algorithms used result in differences in computational cost. For instance, the algorithms **IdealToIsogeny**, **GenRandIsogImg**, and **GenImRanIso**, employed in the computation of φ_{aux} in each scheme, require approximately 4λ , 3λ , and 2λ times $(2, 2)$ -isogeny computations, respectively, under λ -bit security.

3 New Algorithm: PushRandIsog

In this section, we propose a new algorithm named **PushRandIsog**, which is used to compute an auxiliary isogeny in our new scheme SQIsign2DPush. From now on, we use the following notation.

- A cryptographic size prime $p = c \cdot 2^{e_1} \cdot 3^{e_2} - 1$ with $e_1, e_2 \in \mathbb{N}$ and $c \in \mathbb{N}$ as small as possible.
- The elliptic curve $E_0 : y^2 = x^3 + x$ defined over $\mathbb{F}_{p^2}$.
- A deterministic basis P_0, Q_0 of $E_0[2^{e_1}3^{e_2}]$.
- The maximal order $\mathcal{O}_0 := \mathbb{Z}\langle 1, \mathbf{i}, \frac{1+\mathbf{j}}{2}, \frac{1+\mathbf{k}}{2} \rangle$ of $\mathcal{B}_{p,\infty}$. This order $\mathcal{O}_0$ is isomorphic to $\text{End}(E_0)$ as described in Section 2.3.
- For integers ℓ, m, n , we denote by $(m, n)_\ell$ the set of integers x such that $m < x < n$ and $\gcd(x, \ell) = 1$.
- For an isogeny $\varphi : E \rightarrow F$ and an integer n , we denote by $\varphi|_n$ the restriction of φ to $E[n]$.

In addition, we let $(P_0^1, Q_0^1) := [3^{e_2}](P_0, Q_0)$, which is a basis of $E_0[2^{e_1}]$ and $(P_0^2, Q_0^2) := [2^{e_1}](P_0, Q_0)$, which is a basis of $E_0[3^{e_2}]$.

3.1 Overview of PushRandIsog

Here, we give an overview of our new algorithm **PushRandIsog**. This algorithm takes as input an integer $d \in (0, 2^{e_1})_6$ and a 3^{2e_2} -isogeny $\psi : E_0 \rightarrow E$ and then output an elliptic curve E' and $\varphi|_{2^{e_1}}$ for a d -isogeny $\varphi : E \rightarrow E'$.

From now on, we decompose ψ into two 3^{e_2} -isogenies $\psi_0 : E_0 \rightarrow E_m$ and $\psi_m : E_m \rightarrow E$. Then we can assume that ψ is given by generators of $\ker \psi_0$ and $\ker \psi_m$ since $E_0[3^{e_2}]$ is $\mathbb{F}_{p^2}$ -rational. We denote these generators by K_0 and K_m , respectively. In addition, we let $P_m^1 := \psi_0(P_0^1)$, $Q_m^1 := \psi_0(Q_0^1)$, $P^1 := \psi_m(P_m^1)$, and $Q^1 := \psi_m(Q_m^1)$. Our algorithm can be broadly divided into the following three steps.

- Compute a $d(2^{e'_1} - d)$ -isogeny $\rho_0 \circ \varphi_0 : E_0 \rightarrow E_0''$ for an integer $e'_1 \leq e_1$ such that $2^{e'_1} - d \in (0, 2^{e_1})_3$.
- Compute a push-forward $\rho_m \circ \varphi_m$ of $\rho_0 \circ \varphi_0$ by ψ_0 .
- Compute a push-forward φ of φ_m by ψ_m .

The following is the commutative diagram of isogenies that our algorithm is based on. Note that the symbol in each parenthesis represents the degree of each isogeny.

$$\begin{array}{ccccc}
 E_0 & \xrightarrow[\quad (d) \quad]{\varphi_0} & E'_0 & \xrightarrow[\quad (2^{e'_1} - d) \quad]{\rho_0} & E_0'' \\
 \psi_0 \downarrow & & & & \downarrow \psi_0'' \\
 E_m & \xrightarrow[\quad (d) \quad]{\varphi_m} & E'_m & \xrightarrow[\quad (2^{e'_1} - d) \quad]{\rho_m} & E_m'' \\
 \psi_m \downarrow & & \downarrow \psi_m' & & \\
 E & \xrightarrow[\quad (d) \quad]{\varphi} & E' & &
 \end{array}$$

Algorithm 1 Compute a $d(2^{e'_1} - d)$ -isogeny $\rho_m \circ \varphi_m$

Require: An integer $d \in (0, 2^{e_1})_6$, a generator K_0 of ψ_0 , the integer e'_1 in step (i), and $\delta := e_1 - e'_1$.

Ensure: E''_m and $(\rho_m \circ \varphi_m([2^\delta]P_m^1), \rho_m \circ \varphi_m([2^\delta]Q_m^1))$ for a $d(2^{e'_1} - d)$ -isogeny $\rho_m \circ \varphi_m : E_m \rightarrow E''_m$.

- 1: Let $\alpha \leftarrow \text{FullRepresentInteger}_{\mathcal{O}_0}(d(2^{e'_1} - d)2^\delta 3^{e_2})$.
 // Write $\alpha = \hat{\tau}_0 \circ \rho_0 \circ \varphi_0$, where $\deg \varphi_0 = d$, $\deg \rho_0 = 2^{e'_1} - d$, $\deg \tau_0 = 2^\delta 3^{e_2}$.
 - 2: Let $\tau_0 : E_0 \rightarrow E''_0$ be a $2^\delta 3^{e_2}$ -isogeny with kernel $E_0[2^\delta 3^{e_2}] \cap \ker(\hat{\alpha})$.
 - 3: Let $R''_0, S''_0 \leftarrow \tau_0 \circ \alpha(P_0), \tau_0 \circ \alpha(Q_0)$.
 // $R''_0 = \rho_0 \circ \varphi_0([2^\delta P_0^1])$, $S''_0 = \rho_0 \circ \varphi_0([2^\delta Q_0^1])$.
 - 4: Let P''_0, Q''_0 be a basis of $E''_0[3^{e_2}]$.
 - 5: $(E'_0; K'_0; P'_0, Q'_0) \leftarrow \text{KaniEval}(d, 2^{e'_1} - d, [2^\delta]P_0^1, [2^\delta]Q_0^1, R''_0, S''_0; K_0; P''_0, Q''_0)$.
 // $K'_0 = \varphi_0(K_0)$, $P'_0 = \hat{\rho}_0(P''_0)$, $Q'_0 = \hat{\rho}_0(Q''_0)$.
 - 6: Find two integers $u, v \in \mathbb{Z}/3^{e_2}\mathbb{Z}$ such that $K'_0 = [u]P'_0 + [v]Q'_0$. //Solve BiDLP.
 - 7: Let $K''_0 \leftarrow [2^{e'_1} - d]([u]P''_0 + [v]Q''_0)$.
 - 8: Compute a 3^{e_2} -isogeny $\psi''_0 : E''_0 \rightarrow E''_m$ with kernel $\langle K''_0 \rangle$ and let $R''_m \leftarrow \psi''_0(R''_0)$, $S''_m \leftarrow \psi''_0(S''_0)$.
 - 9: **return** $E''_m, (R''_m, S''_m)$.
-

An integer $e'_1 \leq e_1$ such that $2^{e'_1} - d \in (0, 2^{e_1})_3$ might not exist in general (e.g. when $d = 2^{e_1} - 3$), but we here assume that there exists such an integer e'_1 . Note that in our protocol setting, we can always find such e'_1 . We will discuss it in Remark 1 of Section 4.

3.2 Details of PushRandIsog

Now we show the details of **PushRandIsog**. First, the computations related to the upper half of the commutative diagram (namely steps (i) and (ii)) are performed in the same manner as the key generation and encryption in POKE [5, Algorithm 3, Algorithm 4]. We show the concrete algorithm for steps (i) and (ii) in Algorithm 1.

We show that the points (R''_m, S''_m) in the output of Algorithm 1 are equal to $(\rho_m \circ \varphi_m([2^\delta]P_m^1), \rho_m \circ \varphi_m([2^\delta]Q_m^1))$. In step 6 in Algorithm 1, we obtain integers u, v such that $\varphi_0(K_0) = [u]\hat{\rho}_0(P''_0) + [v]\hat{\rho}_0(Q''_0) = \hat{\rho}_0([u]P''_0 + [v]Q''_0)$ hold. Then we obtain $\rho_0 \circ \varphi_0(K_0) = [\deg \rho_0]([u]P''_0 + [v]Q''_0) = K''_0$, which is a generator of ψ''_0 . From the commutativity of the upper half of the diagram, $R''_m = \psi''_0(R''_0)$ satisfies the following equation:

$$\psi''_0(R''_0) = \psi''_0 \circ \rho_0 \circ \varphi_0([2^\delta]P_0^1) = \rho_m \circ \varphi_m \circ \psi_0([2^\delta]P_0^1) = \rho_m \circ \varphi_m([2^\delta]P_m^1).$$

Similarly, $S''_m = \rho_m \circ \varphi_m([2^\delta]Q_m^1)$ also holds. This means that we obtain $\rho_m \circ \varphi_m|_{2^{e'_1}}$ since $([2^\delta]P_m^1, [2^\delta]Q_m^1)$ is a basis of $E_m[2^{e'_1}]$.

Unlike POKE, we additionally need to compute the push-forward φ of the isogeny φ_m by ψ_m (step (iii)). We can compute the push-forward φ as follows:

1. Compute $R'_m = \varphi_m(P_m^1)$, $S'_m = \varphi_m(Q_m^1)$, and $K'_m = \varphi_m(K_m)$ from $\rho_m \circ \varphi_m|_{2^{e'_1}}$ via **KaniEval**.

Algorithm 2 PushRandIsog_{O₀}(d, ψ)**Require:** An integer $d \in (0, 2^{e_1})_6$ and a 3^{2e_2} -isogeny $\psi : E_0 \rightarrow E$.**Ensure:** $(E'; \varphi(P^1), \varphi(Q^1))$ for a d -isogeny $\varphi : E \rightarrow E'$ and two points $(P^1, Q^1) = (\psi(P_0^1), \psi(Q_0^1))$.

- 1: Obtain an elliptic curve E_m and points $K_0 \in E_0, K_m \in E_m$ such that ψ is the composition of two 3^{e_2} -isogenies $\psi_0 : E_0 \rightarrow E_m$ with kernel $\langle K_0 \rangle$ and $\psi_m : E_m \rightarrow E$ with kernel $\langle K_m \rangle$.
- 2: Let $P_m^1, Q_m^1 \leftarrow \psi_0(P_0^1), \psi_0(Q_0^1)$.
- 3: Find the smallest integer $e'_1 \leq e_1$ such that $2^{e'_1} - d \in (0, 2^{e_1})_3$ and let $\delta \leftarrow e_1 - e'_1$.
- 4: Compute E_m'' and $(R_m'', S_m'') = (\rho_0 \circ \varphi_0([2^\delta]P_m^1), \rho_0 \circ \varphi_0([2^\delta]Q_m^1))$ for a $d(2^{e'_1} - d)$ -isogeny $\rho_m \circ \varphi_m : E_m \rightarrow E_m''$ via Algorithm 1.
- 5: $(E'_m; R'_m, S'_m, K'_m; \emptyset)$
 $\leftarrow \text{KaniEval}(d, 2^{e'_1} - d, [2^\delta]P_m^1, [2^\delta]Q_m^1, R_m'', S_m'', P_m^1, Q_m^1, K_m; \emptyset).$
 $// R'_m = \varphi_m(P_m^1), S'_m = \varphi_m(Q_m^1), K'_m = \varphi_m(K_m).$
- 6: Compute the 3^{e_2} -isogeny $\psi'_m : E'_m \rightarrow E'$ with kernel $\langle K'_m \rangle$ and let $R' \leftarrow \psi'_m(R'_m), S' \leftarrow \psi'_m(S'_m).$
- 7: **return** $(E'; R', S').$

2. Compute a 3^{e_2} -isogeny $\psi'_m : E'_m \rightarrow E'$ with kernel $\langle K'_m \rangle$.
3. Let $R' \leftarrow \psi'_m(R'_m)$ and $S' \leftarrow \psi'_m(S'_m).$

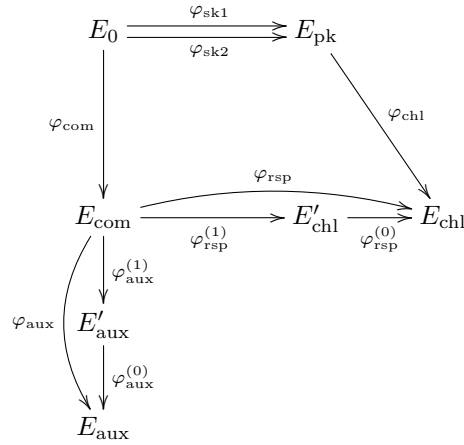
From the commutativity of the lower half of the diagram, we have $R' = \psi'_m \circ \varphi_m(P_m^1) = \varphi \circ \psi_m(P_m^1) = \varphi(P^1)$ and similarly, $S' = \varphi(Q^1)$. We show the detailed algorithm of PushRandIsog in Algorithm 2.

4 New Signature Scheme: SQIsign2DPPush

In this section, we describe our new signature scheme SQIsign2DPPush.

4.1 SQIsign2DPPush Σ -Protocol

We first describe the Σ -protocol underlying SQIsign2DPPush. The Σ -protocol is based on the following diagram.



Parameter setting. We use the same notation as in Section 3, assuming that $2^{e_1} \approx 3^{e_2} \approx 2^\lambda$, and $2^{e_1} \geq 2\sqrt{2p}/\pi$. Then the public parameters of SQIsign2DPush are taken as $\text{param} = (p, e_1, e_2, E_0, P_0, Q_0, \mathcal{O}_0)$. Note that these parameters are the same as SQIsignHD.

Key generation, commitment, and challenge. The algorithms for key generation, commitment, and challenge are almost the same as SQIsignHD.

- **Key generation.** We define $\text{gen}(1^\lambda) \rightarrow (\text{pk}, \text{sk})$ as follows: Compute a 2^{2e_1} -isogeny $\varphi_{\text{sk}1} : E_0 \rightarrow E_{\text{pk}}$, a 3^{2e_2} -isogeny $\varphi_{\text{sk}2} : E_0 \rightarrow E_{\text{pk}}$, and their corresponding ideals $I_{\text{sk}1}$, and $I_{\text{sk}2}$ using **FastDoublePath** $_{2^{e_1}, 3^{e_2}}$. Return $\text{pk} = E_{\text{pk}}$ and $\text{sk} = (\varphi_{\text{sk}1}, \varphi_{\text{sk}2}, I_{\text{sk}1}, I_{\text{sk}2})$.
- **Commitment.** We define $\text{com}(\text{pk}, \text{sk}) \rightarrow (\text{com}, \text{st})$ as follows: Compute a 3^{2e_2} -isogeny $\varphi_{\text{com}} : E_0 \rightarrow E_{\text{com}}$ and its corresponding ideal I_{com} via **FastCommit** $_{2^{e_1}, 3^{e_2}}$. Let ψ_0 and ψ_m be the isogenies such that $\varphi_{\text{com}} = \psi_m \circ \psi_0$ and $\deg \psi_0 = \deg \psi_m = 3^{e_2}$. Return $\text{com} = E_{\text{com}}$ and a state $\text{st} = (\varphi_{\text{com}}, I_{\text{com}})$. Since we use **FastCommit**, we have generators K_0 of $\ker \psi_0$ and K_m of $\ker \psi_m$ as an efficient representation of φ_{com} .
- **Challenge.** We use the challenge space $\mathcal{C} = (0, 3^{e_2})$. A challenge $\text{chl} \xleftarrow{\$} \mathcal{C}$ describes the kernel of a challenge isogeny $\varphi_{\text{chl}} : E_{\text{pk}} \rightarrow E_{\text{chl}}$ such that $\ker(\varphi_{\text{chl}}) = \langle P_{\text{pk}} + [\text{chl}]Q_{\text{pk}} \rangle$, where $P_{\text{pk}}, Q_{\text{pk}}$ is the deterministic basis of $E_{\text{pk}}[3^{e_2}]$.

Response. As in SQIsign-v2.0, we compute a response isogeny $\varphi_{\text{rsp}} : E_{\text{com}} \rightarrow E_{\text{chl}}$ and an auxiliary isogeny $\varphi_{\text{aux}} : E_{\text{com}} \rightarrow E_{\text{aux}}$.

- **rsp**($\text{pk}, \text{sk}, \text{st}, \text{chl}$) $\rightarrow \text{rsp} / \perp$:
 1. Compute the ideal I_{chl} corresponding to φ_{chl} via the isomorphism induced by $\varphi_{\text{sk}1}$ using **IsogToIdeal**($\varphi_{\text{chl}}, \varphi_{\text{sk}1}, I_{\text{sk}1}$).
 2. Let $\gamma \in \mathcal{O}_0$ be the quaternion corresponding to $\widehat{\varphi}_{\text{sk}2} \circ \varphi_{\text{sk}1}$ and let $I = \bar{I}_{\text{com}} \cdot I_{\text{sk}1} \cdot I_{\text{chl}} \cdot \frac{\gamma}{2^{2e_1}}$, which corresponds to $\varphi_{\text{chl}} \circ \varphi_{\text{sk}2} \circ \widehat{\varphi}_{\text{com}}$ via the isomorphism induced by φ_{com} .
 3. Sample an ideal I_{rsp} equivalent to I of norm $\in (0, 2^{e_1})_3$, i.e., let $I_{\text{rsp}} = I \cdot \frac{\bar{\alpha}}{n(I)}$ for $\alpha \in I$ such that $q_I(\alpha) \in (0, 2^{e_1})_3$. If there is no such ideal, return $\perp$, which means a failure of response.
 4. Since $\alpha \in I \subseteq \mathcal{O}_0$, we can consider the endomorphism of E_0 corresponding to α . We denote it by α_{E_0} . Let φ_{rsp} be the isogeny corresponding to I_{rsp} via the isomorphism induced by φ_{com} . Then we can compute $\varphi_{\text{rsp}}|_{2^{e_1}}$ using the equation:

$$\varphi_{\text{rsp}} = [3^{-5e_2}] \circ \varphi_{\text{chl}} \circ \varphi_{\text{sk}2} \circ \hat{\alpha}_{E_0} \circ \hat{\varphi}_{\text{com}}$$

since $\alpha_{E_0} = \widehat{\varphi}_{\text{com}} \circ \widehat{\varphi}_{\text{rsp}} \circ \varphi_{\text{chl}} \circ \varphi_{\text{sk}2}$.

5. Let d_{rsp} be the norm of I_{rsp} and write $d_{\text{rsp}} = d'_{\text{rsp}} \cdot 2^{n_1}$ for an odd integer d'_{rsp} . We decompose φ_{rsp} into $\varphi_{\text{rsp}}^{(0)} \circ \varphi_{\text{rsp}}^{(1)}$ such that $\deg(\varphi_{\text{rsp}}^{(1)}) = d'_{\text{rsp}}$ and $\deg(\varphi_{\text{rsp}}^{(0)}) = 2^{n_1}$. We denote the codomain of $\varphi_{\text{rsp}}^{(1)}$ by E'_{chl} .

6. Let $d_{\text{aux}} = 2^{e_1 - n_1} - d'_{\text{rsp}}$ and write $d_{\text{aux}} = d'_{\text{aux}} \cdot 3^{n_2}$ for an integer d'_{aux} coprime to 3.
7. Since $d'_{\text{aux}} \in (0, 2^{e_1})_6$, we can compute $\varphi_{\text{aux}}^{(1)}|_{2^{e_1}}$ for a d'_{aux} -isogeny $\varphi_{\text{aux}}^{(1)} : E_{\text{com}} \rightarrow E'_{\text{aux}}$ via **PushRandIsog**.
8. If n_2 is odd, compute a random 3-isogeny φ_3 from E'_{aux} and let $\varphi_{\text{aux}}^{(0)} = [3^{(n_2-1)/2}] \circ \varphi_3$. Otherwise, let $\varphi_{\text{aux}}^{(0)} = [3^{n_2/2}]$. Let E_{aux} be the codomain of $\varphi_{\text{aux}}^{(0)}$.
9. Let $\varphi_{\text{aux}} = \varphi_{\text{aux}}^{(0)} \circ \varphi_{\text{aux}}^{(1)}$, whose degree is d_{aux} .
10. Take the deterministic basis $P_{\text{aux}}, Q_{\text{aux}}$ of $E_{\text{aux}}[2^{e_1}]$.
11. Let $(P_{\text{chl}}, Q_{\text{chl}}) \leftarrow [d_{\text{aux}}^{-1}](\varphi_{\text{rsp}} \circ \widehat{\varphi}_{\text{aux}}(P_{\text{aux}}), \varphi_{\text{rsp}} \circ \widehat{\varphi}_{\text{aux}}(Q_{\text{aux}}))$.
12. Return $\text{rsp} = (E_{\text{aux}}, P_{\text{chl}}, Q_{\text{chl}}, n_1)$.

The response algorithm is shown in Algorithm 3. In step 13 of Algorithm 3, we solve BiDLP on E_{aux} twice to find the matrix M . Since $(R_{\text{aux}}, S_{\text{aux}}) = (\varphi_{\text{aux}}(P_{\text{com}}^1), \varphi_{\text{aux}}(Q_{\text{com}}^1))$, we have

$$(P_{\text{aux}}, Q_{\text{aux}}) = (\varphi_{\text{aux}}(P_{\text{com}}^1), \varphi_{\text{aux}}(Q_{\text{com}}^1))M.$$

By taking $\varphi_{\text{rsp}} \circ \widehat{\varphi}_{\text{aux}}$ on both side, we have

$$\begin{aligned} (\varphi_{\text{rsp}} \circ \widehat{\varphi}_{\text{aux}}(P_{\text{aux}}), \varphi_{\text{rsp}} \circ \widehat{\varphi}_{\text{aux}}(Q_{\text{aux}})) &= [d_{\text{aux}}](\varphi_{\text{rsp}}(P_{\text{com}}^1), \varphi_{\text{rsp}}(Q_{\text{com}}^1))M \\ &= [d_{\text{aux}}](R_{\text{chl}}, S_{\text{chl}})M. \end{aligned}$$

Therefore, in step 14, we obtain

$$(P_{\text{chl}}, Q_{\text{chl}}) = [d_{\text{aux}}^{-1}](\varphi_{\text{rsp}} \circ \widehat{\varphi}_{\text{aux}}(P_{\text{aux}}), \varphi_{\text{rsp}} \circ \widehat{\varphi}_{\text{aux}}(Q_{\text{aux}})).$$

Remark 1. In **PushRandIsog** in step 10 of Algorithm 3, we have to find an integer $e'_1 \leq e_1$ such that $2^{e'_1} - d'_{\text{aux}} \in (0, 2^{e_1})_3$ (see Algorithm 2). We show that we can always find such an integer. (i) If $n_2 = 0$: $d'_{\text{aux}} = 2^{e_1 - n_1} - d'_{\text{rsp}}$ holds and we have $2^{e_1 - n_1} - d'_{\text{aux}} = d'_{\text{rsp}} > 0$. Since $d_{\text{rsp}} = d'_{\text{rsp}} \cdot 2^{n_1}$ is coprime to 3, d'_{rsp} is also coprime to 3. Therefore, we can choose $e'_1 = e_1 - n_1$. (ii) If $n_2 \geq 1$: we have $d'_{\text{aux}} = (2^{e_1 - n_1} - d'_{\text{rsp}})/3^{n_2} < (2^{e_1 - n_1} - d'_{\text{rsp}})/2 < 2^{e_1 - n_1 - 1}$. Hence, $2^{e_1 - n_1} - d'_{\text{aux}} > 2^{e_1 - n_1 - 1} - d'_{\text{aux}} > 0$ holds. Since $2^{e_1 - n_1} - d'_{\text{aux}} \not\equiv 2^{e_1 - n_1 - 1} - d'_{\text{aux}} \pmod{3}$, at least one of them is coprime to 3. Therefore, we can choose $e'_1 = e_1 - n_1$ or $e'_1 = e_1 - n_1 - 1$.

Verify. The verification is also similar to SQIsign-v2.0. But we have to check that $\ker(\widehat{\varphi}_{\text{rsp}} \circ \varphi_{\text{chl}})$ does not contain $E_{\text{pk}}[3]$. This is to achieve the special soundness (see the proof of Theorem 2 for details).

- **verf**(pk, com, chl, rsp) $\rightarrow$ accept/reject: We first compute the isogeny $\varphi_{\text{chl}} : E_{\text{pk}} \rightarrow E_{\text{chl}}$ with kernel $\langle P_{\text{pk}} + [\text{chl}]Q_{\text{pk}} \rangle$ and let K_{bt} be a generator of $\ker \widehat{\varphi}_{\text{chl}} \cap E_{\text{chl}}[3]$, which is the kernel of the first step of the dual $\widehat{\varphi}_{\text{chl}}$. (See Remark 2 for how to compute such K_{bt} .) Next, we compute the isogeny $\varphi : E_{\text{chl}} \rightarrow E'_{\text{chl}}$ with kernel $[2^{e_1 - n_1}]\langle P_{\text{chl}}, Q_{\text{chl}} \rangle$, which corresponds to the

Algorithm 3 $\text{rsp}(\text{pk}, \text{sk}, \text{st}, \text{chl}) \rightarrow \text{rsp} / \perp$ **Require:** A public key pk , a secret key sk , a state st , and a challenge chl .**Ensure:** Response rsp or $\perp$.

- 1: Let $I_{\text{chl}} \leftarrow \text{IsogToIdeal}(\varphi_{\text{chl}}, \varphi_{\text{sk1}}, I_{\text{sk1}})$.
- 2: Let $\gamma \in \mathcal{O}_0$ be the quaternion corresponding to $\widehat{\varphi}_{\text{sk2}} \circ \varphi_{\text{sk1}}$.
- 3: Let $I \leftarrow \bar{I}_{\text{com}} \cdot I_{\text{sk1}} \cdot I_{\text{chl}} \cdot \frac{\gamma}{2^{2e_1}}$.
- 4: Find all $\alpha \in I$ such that $d_{\text{rsp}} := q_I(\alpha) \in (0, 2^{e_1})_3$ by lattice enumeration and choose one of them uniformly. If there is no such α , **return** $\perp$.
- 5: Let $I_{\text{rsp}} = I \frac{\bar{\alpha}}{n(I)}$ and let $\varphi_{\text{rsp}} : E_{\text{com}} \rightarrow E_{\text{chl}}$ be its corresponding isogeny.
- 6: Let $R_{\text{chl}} \leftarrow [3^{-5e_2}] \varphi_{\text{chl}} \circ \varphi_{\text{sk2}} \circ \hat{\alpha}_{E_0} \circ \hat{\varphi}_{\text{com}}(P_{\text{com}}^1)$. $// R_{\text{chl}} = \varphi_{\text{rsp}}(P_{\text{com}}^1)$.
- 7: Let $S_{\text{chl}} \leftarrow [3^{-5e_2}] \varphi_{\text{chl}} \circ \varphi_{\text{sk2}} \circ \hat{\alpha}_{E_0} \circ \hat{\varphi}_{\text{com}}(Q_{\text{com}}^1)$. $// S_{\text{chl}} = \varphi_{\text{rsp}}(Q_{\text{com}}^1)$.
- 8: Let $d_{\text{rsp}} = d'_{\text{rsp}} \cdot 2^{n_1}$, where $2 \nmid d'_{\text{rsp}}$ and let $d_{\text{aux}} = 2^{e_1 - n_1} - d'_{\text{rsp}}$.
- 9: Let $d_{\text{aux}} = d'_{\text{aux}} \cdot 3^{n_2}$, where $3 \nmid d'_{\text{aux}}$.
- 10: Let $(E'_{\text{aux}}, R'_{\text{aux}}, S'_{\text{aux}}) \leftarrow \text{PushRandIsog}_{\mathcal{O}_0}(d'_{\text{aux}}, \varphi_{\text{com}})$.
 $// R'_{\text{aux}} = \varphi_{\text{aux}}^{(1)}(P_{\text{com}}^1), S'_{\text{aux}} = \varphi_{\text{aux}}^{(1)}(Q_{\text{com}}^1)$.
- 11: If n_2 is odd, compute a random 3-isogeny $\varphi_3 : E'_{\text{aux}} \rightarrow E_{\text{aux}}$ and let $\varphi_{\text{aux}}^{(0)} = [3^{(n_2-1)/2}] \circ \varphi_3$. Otherwise, let $\varphi_{\text{aux}}^{(0)} = [3^{n_2/2}]$.
- 12: Let $R_{\text{aux}}, S_{\text{aux}} \leftarrow \varphi_{\text{aux}}^{(0)}(R'_{\text{aux}}), \varphi_{\text{aux}}^{(0)}(S'_{\text{aux}})$.
- 13: Let $P_{\text{aux}}, Q_{\text{aux}}$ be the deterministic basis of $E_{\text{aux}}[2^{e_1}]$ and find a matrix $M \in (\mathbb{Z}/2^{e_1}\mathbb{Z})^{2 \times 2}$ such that $(P_{\text{aux}}, Q_{\text{aux}}) = (R_{\text{aux}}, S_{\text{aux}})M$.
- 14: Let $(P_{\text{chl}}, Q_{\text{chl}}) \leftarrow (R_{\text{chl}}, S_{\text{chl}})M$.
- 15: **return** $(E_{\text{aux}}, P_{\text{chl}}, Q_{\text{chl}}, n_1)$.

dual of $\varphi_{\text{rsp}}^{(0)}$ and let $P'_{\text{chl}} = \varphi(P_{\text{chl}}), Q'_{\text{chl}} = \varphi(Q_{\text{chl}})$, and $K'_{\text{bt}} = \varphi(K_{\text{bt}})$. Then we compute the isogeny Φ with kernel

$$\begin{aligned} & \langle ([2^{n_1}]P_{\text{aux}}, P'_{\text{chl}}), ([2^{n_1}]Q_{\text{aux}}, Q'_{\text{chl}}) \rangle \\ &= [2^{n_1}] \langle ([d_{\text{aux}}]P_{\text{aux}}, \varphi_{\text{rsp}}^{(1)} \circ \widehat{\varphi}_{\text{aux}}(P_{\text{aux}})), ([d_{\text{aux}}]Q_{\text{aux}}, \varphi_{\text{rsp}}^{(1)} \circ \widehat{\varphi}_{\text{aux}}(Q_{\text{aux}})) \rangle. \end{aligned}$$

From Theorem 1, Φ should be a $(2^{e_1 - n_1}, 2^{e_1 - n_1})$ -isogeny from $E_{\text{aux}} \times E'_{\text{chl}}$ to $E_{\text{com}} \times F$ for a curve F . Otherwise, we return reject. Finally, we check whether the kernel of $\widehat{\varphi}_{\text{chl}} \circ \varphi_{\text{rsp}}$ contains $E_{\text{com}}[3]$: compute $K_{\text{com}} = \widehat{\varphi}_{\text{rsp}}^{(1)}(K'_{\text{bt}}) = \widehat{\varphi}_{\text{rsp}}(K_{\text{bt}})$ and check if $K_{\text{com}} = O$. If $K_{\text{com}} = O$, return reject. Otherwise, return accept.

We show the verification algorithm in Algorithm 4.

Remark 2. We explain how to compute K_{bt} in the verification. Let $\varphi : E_1 \rightarrow E_2$ be a 3-isogeny. Then we can compute a generator K_{bt} of $\ker \widehat{\varphi}$ from the Montgomery coefficients of E_1 and E_2 . Let A_1 and A_2 be the Montgomery coefficients of E_1 and E_2 , respectively. From the formulas in [11, Theorem 1], the x -coordinate x_{bt} of K_{bt} satisfies $6x_{\text{bt}} + A_2x_{\text{bt}}^2 - 6x_{\text{bt}}^3 = A_1$. In addition, from the 3-division polynomial of Montgomery curves, we have $3x_{\text{bt}}^4 + 4A_2x_{\text{bt}}^3 + 6x_{\text{bt}}^2 - 1 = 0$. From these two equations, we can express x_{bt} as a rational function of A_1 and A_2 .

Algorithm 4 $\mathbf{verf}(\text{pk}, \text{com}, \text{chl}, \text{rsp}) \rightarrow \text{accept/reject}$

Require: A public key pk , a commitment com , a challenge chl , and a response rsp .

Ensure: accept or reject.

- 1: Let $P_{\text{pk}}, Q_{\text{pk}}$ be the deterministic basis of $E_{\text{pk}}[3^{e_2}]$.
 - 2: Compute a 3^{e_2} -isogeny $\varphi_{\text{chl}} : E_{\text{pk}} \rightarrow E_{\text{chl}}$ with kernel $\langle P_{\text{pk}} + [\text{chl}]Q_{\text{pk}} \rangle$.
 - 3: Let K_{bt} be a generator of $\ker \widehat{\varphi}_{\text{chl}} \cap E_{\text{chl}}[3]$.
 - 4: Compute a 2^{n_1} -isogeny $\varphi : E_{\text{chl}} \rightarrow E'_{\text{chl}}$ with kernel $[2^{e_1-n_1}]\langle P_{\text{chl}}, Q_{\text{chl}} \rangle$ and let $P'_{\text{chl}} \leftarrow \varphi(P_{\text{chl}}), Q'_{\text{chl}} \leftarrow \varphi(Q_{\text{chl}}), K'_{\text{bt}} \leftarrow \varphi(K_{\text{bt}})$.
 - 5: Let $P_{\text{aux}}, Q_{\text{aux}}$ be the deterministic basis of $E_{\text{aux}}[2^{e_1}]$.
 - 6: Compute a $(2^{e_1-n_1}, 2^{e_1-n_1})$ -isogeny $\Phi : E_{\text{aux}} \times E'_{\text{chl}} \rightarrow A$ with kernel $\langle ([2^{n_1}]P_{\text{aux}}, P'_{\text{chl}}), ([2^{n_1}]Q_{\text{aux}}, Q'_{\text{chl}}) \rangle$.
 - 7: **if** not $A \cong E_{\text{com}} \times F$ for a curve F **then**
 - 8: **return** reject.
 - 9: **end if**
 - 10: Let $\eta : A \rightarrow E_{\text{com}} \times F$ be the isomorphism and let $(K_{\text{com}}, -) \leftarrow \eta \circ \Phi((O, K'_{\text{bt}}))$.
 - 11: **if** $K_{\text{com}} \neq O$ **then**
 - 12: **return** reject.
 - 13: **end if**
 - 14: **return** accept.
-

Correctness Assume that the response is generated honestly. As we mentioned in the description of **verf**, the map Φ computed in step 6 should be a $(2^{e_1-n_1}, 2^{e_1-n_1})$ -isogeny from $E_{\text{aux}} \times E'_{\text{chl}}$ to $E_{\text{com}} \times F$ for a curve F . In addition, $\widehat{\varphi}_{\text{chl}} \circ \varphi_{\text{rsp}}$ does not contain $E_{\text{com}}[3]$ since φ_{chl} is a cyclic 3^{e_2} -isogeny and $\deg \varphi_{\text{rsp}}$ is coprime to 3. Therefore, **verf** always accepts the honest response.

4.2 Fiat-Shamir with Aborts

To obtain a signature scheme from our Σ -protocol, we apply the Fiat-Shamir transform. However, our Σ -protocol may fail to generate a response, i.e. **rsp** may output $\perp$ when there is no ideal I_{rsp} of desired norm. Therefore, we apply the Fiat-Shamir transform with aborts (FSwA) [29]. Let $H : \{0, 1\}^* \rightarrow [0, 3^{e_2})$ be a cryptographic hash function. Then FSwA is as follows.

Fiat-Shamir transform with aborts

Sign(pk, sk, m):

1. $\text{rsp} \leftarrow \perp$.
2. **while** $\text{rsp} = \perp$:
 - 2-1. $(\text{com}, \text{st}) \leftarrow \mathbf{com}(\text{pk}, \text{sk})$.
 - 2-2. $\text{chl} \leftarrow H(\text{pk}, m, \text{com})$.
 - 2-3. $\text{rsp} \leftarrow \mathbf{rsp}(\text{pk}, \text{sk}, \text{st}, \text{chl})$.
3. **return** $\text{sig} = (\text{com}, \text{rsp})$.

Verf(pk, m, sig):

1. Parse $\text{sig} = (\text{com}, \text{rsp})$.
2. $\text{chl} \leftarrow H(\text{pk}, m, \text{com})$.
3. **return** $\mathbf{verf}(\text{pk}, \text{com}, \text{chl}, \text{rsp})$.

If the failure probability of **rsp** is bounded by a constant $c < 1$, this signing procedure stops in polynomial time. In Appendix C, we show that the failure

probability is heuristically at most $1/3$, theoretically at most $2/3$, and experimentally at most 2^{-18} . In addition, we discuss the effect of the failure probability on the security of the protocol in Section 5.1.

4.3 Signature Encoding and Sizes

The signature of SQIsign2DPush consists of $(E_{\text{com}}, E_{\text{aux}}, P_{\text{chl}}, Q_{\text{chl}}, n_1)$, but the signature size can be compressed similarly to SQIsign-v2.0.

First, two points $P_{\text{chl}}, Q_{\text{chl}}$ can be represented by four elements of $\mathbb{Z}/2^{e_1}\mathbb{Z}$ (that are the coefficients of $P_{\text{chl}}, Q_{\text{chl}}$ on the deterministic basis of $E_{\text{chl}}[2^{e_1}]$). To compute the coefficients, the signer additionally needs to solve BiDLP on $E_{\text{chl}}[2^{e_1}]$ twice.

Moreover, we can reduce the size of the signature by including the challenge $\text{chl} \in [0, 3^{e_2})$ instead of the commitment E_{com} . In that case, the verifier first computes the isogeny φ_{chl} from chl and obtains its codomain E_{chl} . The verifier then computes $\varphi : E_{\text{chl}} \rightarrow E'_{\text{chl}}$ and $\Phi : E_{\text{aux}} \times E'_{\text{chl}} \rightarrow A$ as in steps 1-6 of Algorithm 4. The verifier then checks if $A \cong F_1 \times F_2$ for curves F_1 and F_2 . By computing two hash values $H(\text{pk}, m, j(F_1))$ and $H(\text{pk}, m, j(F_2))$ and comparing them with chl , the verifier can determine whether F_1 or F_2 is isomorphic to the commitment curve E_{com} . After obtaining the curve E_{com} , the verifier performs steps 10-14 of Algorithm 4 in the same manner.

As in SQIsign-v2.0, we include two hints in the signature to allow the verifier to compute the deterministic basis of $E_{\text{chl}}[2^{e_1}]$ and $E_{\text{aux}}[2^{e_1}]$ faster. We use [1, Algorithm 2.1] to generate a hint and use [1, Algorithm 2.2] to compute the basis using the hint. The size of each hint is 1 byte. We also include a hint for the deterministic basis of $E_{\text{pk}}[3^{e_2}]$ in the public key. For this, we use the method of [12, Section 3.3] and [4, Section 7.1]. The hint for $E_{\text{pk}}[3^{e_2}]$ costs 2 bytes.

As a result, the signature of our protocol requires $\lceil \log_2 3^{e_2} \rceil$ bits for chl , $\lceil \log_2 p^2 \rceil$ bits for E_{aux} , $4\lceil \log_2 2^{e_1} \rceil$ bits for $P_{\text{chl}}, Q_{\text{chl}}$, $\lceil \log_2 e_1 \rceil$ bits for n_1 , and 2 bytes for two hints. In total, the signature size is approximately $9\lambda + \log_2 \lambda + 16$ bits since $2^{e_1} \approx 3^{e_2} \approx 2^\lambda$.

5 Security Analysis

In this section, we analyze the security of SQIsign2DPush. Roughly speaking, to ensure the EUF-CMA security, we have to prove that the underlying Σ -protocol is knowledge-sound and honest-verifier zero-knowledge.

In our proof, we employ the Fiat-Shamir with hints (FSwH) framework [2, Section 3], as in SQIsign. However, as described in Section 4.2, our protocol is derived by applying the Fiat-Shamir transform with aborts (FSwA) to our Σ -protocol. Therefore, it is necessary to conduct a new proof within a framework that incorporates both FSwH and FSwA. Therefore, in Section 5.1, we first discuss the EUF-CMA security in general when combining both FSwH and FSwA, and give a new theorem to prove the EUF-CMA security (Theorem 1). Following this, we specifically address the security of SQIsign2DPush.

5.1 Combining FSwH and FSwA frameworks

Let $\Sigma = (P_1, P_2, V)$ be a Σ -protocol against a relation $\mathcal{R} \subset \mathcal{X} \times \mathcal{W}$ with a challenge space $\mathcal{C}$. We define the commitment min-entropy of Σ as follows:

$$\text{MinEnt}(\Sigma) := \max_{(x,w) \in \mathcal{R}} \max_{\text{com}'} \Pr[\text{com} = \text{com}' \mid (\text{com}, \text{st}) \leftarrow P_1(x, w)],$$

which is the same definition as in [2, Definition 2.4]. We denote the maximal failure probability of Σ by

$$\mathcal{P}_\Sigma^\perp := \max_{(x,w) \in \mathcal{R}} \Pr \left[P_2(x, w, \text{st}, \text{chl}) = \perp \mid (\text{com}, \text{st}) \leftarrow P_1(x, w), \text{chl} \xleftarrow{\$} \mathcal{C} \right].$$

We denote the signature obtained by applying Fiat-Shamir transform with aborts (FSwA) to Σ by $\text{SIG}^\perp[\Sigma]$. We first recall several definitions related to FSwH framework.

Definition 1 ([2, Definition 3.1]). A hint distribution $\mathcal{H}$ for $\mathcal{R}$ is a collection of distributions $\{\mathcal{H}_x\}_{x \in \mathcal{X}}$ on a set HintSet_x of hints for each x .

Definition 2 ([2, Definition 3.3]). Let $\mathcal{H}$ be a hint distribution for $\mathcal{R}$. We say that $(\mathcal{R}, \mathcal{H})$ is a hard relation with hints if for all PPT algorithms $\mathcal{A}$ and all $q = \text{poly}(\lambda)$, it holds that

$$\text{Adv}_{(\mathcal{R}, \mathcal{H})}^{\text{hint-rel}}(\mathcal{A}, q) := \Pr \left[(x, w^*) \in \mathcal{R} \mid \begin{array}{l} (x, w) \leftarrow \mathbf{Gen}_{\mathcal{R}}(1^\lambda), \\ h_1, \dots, h_q \leftarrow \mathcal{H}_x, \\ w^* \leftarrow \mathcal{A}(x, h_1, \dots, h_q) \end{array} \right] = \text{negl}(\lambda).$$

Definition 3 ([2, Definition 3.5]). Let $\mathcal{H}$ be a hint distribution for $\mathcal{R}$. We say that Σ is secure under $\mathcal{H}$ -hint-assisted passive impersonation attacks ($\mathcal{H}$ -*hint-IMP-PA*) if for all $q = \text{poly}(\lambda)$ and all PPT two-stage algorithms $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ it holds that

$$\text{Adv}_{\Sigma, \mathcal{H}}^{\text{hint-IMP-PA}}(\mathcal{A}, q) := \Pr[\mathcal{H}\text{-hint-IMP-PA}_\Sigma(\mathcal{A}, q) = 1] = \text{negl}(\lambda),$$

where $\mathcal{H}\text{-hint-IMP-PA}_\Sigma$ is the following game.

Game: $\mathcal{H}\text{-hint-IMP-PA}_\Sigma(\mathcal{A}, q)$

1. $(x, w) \leftarrow \mathbf{Gen}_{\mathcal{R}}(1^\lambda)$
2. $h_1, \dots, h_q \leftarrow \mathcal{H}_x$
3. $(\text{com}^*, \text{st}^*) \leftarrow \mathcal{A}_1(x, h_1, \dots, h_q)$
4. $\text{chl}^* \xleftarrow{\$} \mathcal{C}$
5. $\text{rsp}^* \leftarrow \mathcal{A}_2(\text{st}^*, \text{chl}^*)$
6. **return** $V(x, \text{com}^*, \text{chl}^*, \text{rsp}^*)$

Next, we define several terms for our security analysis.

Definition 4 (IMP-PA[⊥]). We say that Σ is secure under passive impersonation attacks with abort (IMP-PA[⊥]) if for all PPT two-stage algorithms $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, it holds that

$$\text{Adv}_{\Sigma}^{\text{IMP-PA}^{\perp}}(\mathcal{A}) := \Pr[\text{IMP-PA}_{\Sigma}^{\perp}(\mathcal{A}) = 1] = \text{negl}(\lambda),$$

where $\text{IMP-PA}_{\Sigma}^{\perp}$ is the following game.

Game: $\text{IMP-PA}_{\Sigma}^{\perp}(\mathcal{A})$

- | | |
|---|--|
| <ol style="list-style-type: none"> 1. $(x, w) \leftarrow \text{Gen}_{\mathcal{R}}(1^{\lambda})$ 2. $(\text{com}^*, \text{st}^*) \leftarrow \mathcal{A}_1^{\text{OTrans}_{x,w}^{\perp}}(x)$ 3. $\text{chl}^* \xleftarrow{\\$} \mathcal{C}$ 4. $\text{rsp}^* \leftarrow \mathcal{A}_2(\text{st}^*, \text{chl}^*)$ 5. return $V(x, \text{com}^*, \text{chl}^*, \text{rsp}^*)$ | <p style="text-align: center;"><u>$\text{OTrans}_{x,w}^{\perp}$:</u></p> <ol style="list-style-type: none"> 1. $\text{rsp} \leftarrow \perp$ 2. while $\text{rsp} = \perp$: <ol style="list-style-type: none"> 2-1. $(\text{com}, \text{st}) \leftarrow P_1(x, w)$ 2-2. $\text{chl} \xleftarrow{\\$} \mathcal{C}$ 2-3. $\text{rsp} \leftarrow P_2(x, w, \text{st}, \text{chl})$ 3. return $(\text{com}, \text{chl}, \text{rsp})$ |
|---|--|

Definition 5 (Hint-assisted wHVZK[⊥]). Let $\mathcal{H}$ be a hint distribution for $\mathcal{R}$. We say that Σ is $\mathcal{H}$ -hint-assisted wHVZK[⊥] if there exists a PPT algorithm, called the simulator $\mathcal{S}$, such that for all $q = \text{poly}(\lambda)$ and all PPT algorithms $\mathcal{A}$

$$\text{Adv}_{\Sigma, \mathcal{H}, \mathcal{S}}^{\text{hint-wHVZK}^{\perp}}(\mathcal{A}) := \left| \frac{\Pr[1 \leftarrow \mathcal{A}(x, \pi) | (x, \pi) \leftarrow \text{Real}_{\Sigma}^{\perp}(1^{\lambda}, q)]}{\Pr[1 \leftarrow \mathcal{A}(x, \pi) | (x, \pi) \leftarrow \text{HintSim}_{\Sigma}(1^{\lambda}, q, \mathcal{S})]} \right| = \text{negl}(\lambda),$$

where $\text{Real}_{\Sigma}^{\perp}(1^{\lambda}, q)$ and $\text{HintSim}_{\Sigma}(1^{\lambda}, q, \mathcal{S})$ are distributions defined as follows.

Hint-assisted wHVZK[⊥]

- | | |
|--|--|
| <p style="text-align: center;"><u>$\text{Real}_{\Sigma}^{\perp}(1^{\lambda}, q)$:</u></p> <ol style="list-style-type: none"> 1. $(x, w) \leftarrow \text{Gen}_{\mathcal{R}}(1^{\lambda})$ 2. for $i = 1$ to q: <ol style="list-style-type: none"> 2-1. $\pi_i \leftarrow \text{OTrans}_{x,w}^{\perp}()$ 3. return $(x, \pi = (\pi_1, \dots, \pi_q))$ | <p style="text-align: center;"><u>$\text{HintSim}_{\Sigma}(1^{\lambda}, q, \mathcal{S})$:</u></p> <ol style="list-style-type: none"> 1. $(x, w) \leftarrow \text{Gen}_{\mathcal{R}}(1^{\lambda})$ 2. for $i = 1$ to q: <ol style="list-style-type: none"> 2-1. $h_i \leftarrow \mathcal{H}_x$ 2-2. $\pi_i \leftarrow \mathcal{S}(x, h_i)$ 3. return $(x, \pi = (\pi_1, \dots, \pi_q))$ |
|--|--|

Now we have the following two lemmas.

Lemma 1. Let Σ be a Σ -protocol. For any PPT algorithm $\mathcal{A}$ attacking the EUF-CMA of $\text{SIG}^{\perp}[\Sigma]$ and for any integer $B \geq 1$, there exists a PPT two-stage algorithm $\mathcal{B}$ such that

$$\text{Adv}_{\text{SIG}^{\perp}[\Sigma]}^{\text{EUF-CMA}}(\mathcal{A}) \leq (1+q) \cdot \text{Adv}_{\Sigma}^{\text{IMP-PA}^{\perp}}(\mathcal{B}) + (1+q+Bq)Bs \cdot \text{MinEnt}(\Sigma) + s \cdot (\mathcal{P}_{\Sigma}^{\perp})^B,$$

where q and s are upper-bounds on the number of queries that $\mathcal{A}$ makes to the random oracle RO and the signing oracle OSign , respectively. Furthermore, $\mathcal{B}$ makes at most s queries to the oracle $\text{OTrans}^{\perp}$.

Lemma 2. *Let $(\mathcal{R}, \mathcal{H})$ be a hard relation with hints and let Σ be a Σ -protocol against the relation $\mathcal{R}$. In addition, let $\mathcal{S}$ be a $\mathcal{H}$ -assisted simulator for Σ . For any PPT two-stage algorithm $\mathcal{A}$ against $\text{IMP-PA}^\perp$ of Σ , there exists a PPT two-stage algorithm $\mathcal{B}$ and a PPT algorithm $\mathcal{D}$ such that*

$$\text{Adv}_{\Sigma}^{\text{IMP-PA}^\perp}(\mathcal{A}) \leq \text{Adv}_{\Sigma, \mathcal{H}}^{\text{hint-IMP-PA}}(\mathcal{B}, s) + \text{Adv}_{\Sigma, \mathcal{H}, \mathcal{S}}^{\text{hint-wHVZK}^\perp}(\mathcal{D}, s),$$

where s is an upper-bound on the number of queries that $\mathcal{A}$ makes to $\text{OTrans}^\perp$.

The proofs for these lemmas are given in Appendix B.

Remark 3. The right-hand side of the inequality in Lemma 1 depends on B . However, by utilizing an integer that minimizes this side, the expression can be reformulated to exclude B . Nonetheless, expressing such an integer B in a closed form is not feasible.

Combining Lemma 1, 2, and [2, Lemma 3.2], we obtain the following theorem.

Theorem 1. *Let $(\mathcal{R}, \mathcal{H})$ be a relation with hints and let Σ be a Σ -protocol against the relation $\mathcal{R}$ that has challenge space $\mathcal{C}$ and is special-sound with respect to the relation $\mathcal{R}$. In addition, let $\mathcal{S}$ be a $\mathcal{H}$ -assisted simulator for Σ . For any PPT algorithm $\mathcal{A}$ attacking the EUF-CMA of $\text{SIG}^\perp[\Sigma]$ and for any integer $B \geq 1$, there exists a PPT algorithm $\mathcal{B}$ and an expected polynomial time algorithm $\mathcal{D}$ such that*

$$\begin{aligned} \text{Adv}_{\text{SIG}^\perp[\Sigma]}^{\text{EUF-CMA}}(\mathcal{A}) \leq & (1+q) \left(\text{Adv}_{(\mathcal{R}, \mathcal{H})}^{\text{hint-rel}}(\mathcal{B}, s) + \text{Adv}_{\Sigma, \mathcal{H}, \mathcal{S}}^{\text{hint-wHVZK}^\perp}(\mathcal{D}, s) + \frac{1}{\#\mathcal{C}} \right) \\ & + (1+q+Bs)Bs \cdot \text{MinEnt}(\Sigma) + s \cdot (\mathcal{P}_{\Sigma}^\perp)^B, \end{aligned}$$

where q and s are upper-bounds on the number of queries that $\mathcal{A}$ makes to RO and OSign , respectively.

Therefore, to prove that $\text{SIG}^\perp[\Sigma]$ is EUF-CMA secure, it is sufficient to prove the following for a hard relation with hints $(\mathcal{R}, \mathcal{H})$.

1. Σ is special sound with respect to the relation $\mathcal{R}$.
2. Σ is hint-assisted wHVZK $^\perp$.
3. Σ has a challenge space of exponential size in λ .
4. Σ has high commitment entropy.
5. The failure probability $\mathcal{P}_{\Sigma}^\perp$ is bounded by a constant $c < 1$.

In our case for SQIsign2DPush, the third condition is obvious. In addition, the forth condition follows from the following heuristic about the output distribution of **FastCommit**, which is discussed in [13, Section 5.2].

Heuristic 1 *The commitment curve E_{com} computed by **FastCommit** is computationally indistinguishable from an elliptic curve chosen uniformly at random from the set of all supersingular elliptic curves over $\mathbb{F}_{p^2}$ up to isomorphism.*

As for the fifth condition, we show that $\mathcal{P}_{\Sigma}^\perp$ is bounded by $1/3$ under Heuristic 2 in Appendix C.1. In the following, we prove that our Σ -protocol is special sound and hint-assisted wHVZK $^\perp$ for a hard relation with hints $(\mathcal{R}, \mathcal{H})$.

5.2 Special Soundness

From now on, we denote our Σ -protocol by Σ_{Push} . We first show that Σ_{Push} is special-sound with respect to the following relation:

$$\mathcal{R}_{\text{OneEnd}} = \{(E, \alpha) \mid E : \text{supersingular curve over } \mathbb{F}_{p^2}, \alpha \in \text{End}(E) \setminus \mathbb{Z}\}.$$

Theorem 2 Σ_{Push} satisfies the special-soundness with respect to the relation $\mathcal{R}_{\text{OneEnd}}$.

Proof. The proof of the soundness of our protocol is similar to that of SQIsignHD [13, Proposition 17]. Let $(E_{\text{com}}, \text{chl}, \text{rsp} = (E_{\text{aux}}, P_{\text{chl}}, Q_{\text{chl}}, n_1))$ and $(E_{\text{com}}, \widetilde{\text{chl}}, \widetilde{\text{rsp}} = (\widetilde{E}_{\text{aux}}, \widetilde{P}_{\text{chl}}, \widetilde{Q}_{\text{chl}}, \widetilde{n}_1))$ be two Σ_{Push} transcripts against the statement E_{pk} with the same commitment E_{com} but different challenges $\text{chl} \neq \widetilde{\text{chl}}$. From chl and $\widetilde{\text{chl}}$, we can compute $\varphi_{\text{chl}} : E_{\text{pk}} \rightarrow E_{\text{chl}}$ and $\widetilde{\varphi}_{\text{chl}} : E_{\text{pk}} \rightarrow \widetilde{E}_{\text{chl}}$. From rsp and $\widetilde{\text{rsp}}$, we can compute efficient representations of $\varphi_{\text{rsp}} : E_{\text{com}} \rightarrow E_{\text{chl}}$ and $\widetilde{\varphi}_{\text{rsp}} : E_{\text{com}} \rightarrow \widetilde{E}_{\text{chl}}$.

Therefore, we obtain an efficient representation of $\beta = \widehat{\varphi}_{\text{chl}} \circ \varphi_{\text{rsp}} \circ \widehat{\widetilde{\varphi}}_{\text{rsp}} \circ \widetilde{\varphi}_{\text{chl}} \in \text{End}(E_{\text{pk}})$. We now show that β is non-scalar. If it were, we would have $\beta = [\nu]$ for an integer ν . Then we have $\nu^2 = \deg \beta = d\tilde{d} \cdot 3^{2e_2}$, where $d = \deg \varphi_{\text{rsp}}$ and $\tilde{d} = \deg \widetilde{\varphi}_{\text{rsp}}$. We write $d = 3^m \cdot d'$ and $\tilde{d} = 3^n \cdot \tilde{d}'$, where d' and $\tilde{d}'$ are coprime to 3. Without loss of generality, we can assume $m \geq n$. Since $\nu^2 = d\tilde{d} \cdot 3^{2e_2} = d' \tilde{d}' \cdot 3^{m+n+2e_2}$, we have $\nu = \nu' \cdot 3^{(m+n)/2+e_2}$ for $\nu' = \sqrt{d' \tilde{d}'} \in \mathbb{Z}$. Note that ν' is coprime to 3 since d' and $\tilde{d}'$ is coprime to 3. Now we have $[\nu'] \widehat{\varphi}_{\text{rsp}} \circ \varphi_{\text{chl}} = [3^{(m-n)/2} \cdot d'] \widehat{\varphi}_{\text{rsp}} \circ \widetilde{\varphi}_{\text{chl}}$. Let $\widetilde{K}_{\text{chl}}$ be a generator of $\ker \widetilde{\varphi}_{\text{chl}}$. Then $\widehat{\varphi}_{\text{rsp}} \circ \varphi_{\text{chl}}(\widetilde{K}_{\text{chl}}) = [(\nu')^{-1} \cdot 3^{(m-n)/2} \cdot d'] \widehat{\varphi}_{\text{rsp}} \circ \widetilde{\varphi}_{\text{chl}}(\widetilde{K}_{\text{chl}}) = O$ holds. Since $\ker(\widehat{\varphi}_{\text{rsp}} \circ \varphi_{\text{chl}})$ does not contain $E_{\text{chl}}[3]$, we have $\varphi_{\text{chl}}(\widetilde{K}_{\text{chl}}) = O$. This means that $\ker \varphi_{\text{chl}} = \ker \widetilde{\varphi}_{\text{chl}}$. This contradicts $\text{chl} \neq \widetilde{\text{chl}}$. $\square$

5.3 Hint-Assisted wHVZK[⊥]

We now show that Σ_{Push} is hint-assisted wHVZK[⊥] (Definition 5). We first give some oracles to define the hint distribution in the following.

$D_{\text{chl}}(E) \rightarrow (\text{chl}, \varphi)$:

Input: A supersingular elliptic curve E over $\mathbb{F}_{p^2}$.

Output: An integer $\text{chl} \in [0, 3^{e_2})$ and a 3^{e_2} -isogeny φ from E .

1. $\text{chl} \xleftarrow{\$} [0, 3^{e_2})$.
2. Take the deterministic basis P, Q of $E[3^{e_2}]$.
3. Compute the isogeny $\varphi : E \rightarrow E'$ with $\ker \varphi = \langle P + [\text{chl}]Q \rangle$.
4. **return** (chl, φ) .

$D_{\text{RspIsog}}(E, E') \rightarrow \varphi / \perp$:

Input: Supersingular elliptic curves E, E' over $\mathbb{F}_{p^2}$.

Output: An isogeny $\varphi : E \rightarrow E'$ of degree $\in (0, 2^{e_1})_3$ or $\perp$.

1. Try to sample an isogeny $\varphi' : E \rightarrow E'$ uniformly among isogenies $E \rightarrow E'$ of degree $\in (0, 2^{e_1})_3$.
2. If there is no such isogeny, **return** $\perp$.
3. Otherwise, write $\varphi' = [m] \circ \varphi$ with $m \in \mathbb{Z}$ and φ is cyclic.
4. **return** φ .

$D_{\text{AuxIsog}}(E, \psi, d) \rightarrow \varphi$:

Input: A supersingular elliptic curve E over $\mathbb{F}_{p^2}$, a 3^{2e_2} -isogeny $\psi : E_0 \rightarrow E$, and an integer $d \in (0, 2^{e_1})_2$.

Output: A d -isogeny φ from E .

1. Write $d = 3^n d'$, where d' coprime to 3.
2. Compute a d' -isogeny $\varphi' : E \rightarrow E'$ via $\text{PushRandIsog}_{\mathcal{O}_0}(d', \psi)$.
3. If n is odd, compute a 3-isogeny $\varphi'' : E' \rightarrow E''$ uniformly and let $\varphi'' \leftarrow [3^{(n-1)/2}] \circ \varphi'$. Otherwise, let $\varphi'' = [3^{n/2}]$.
4. Let $\varphi \leftarrow \varphi'' \circ \varphi'$.
5. **return** φ .

Then we define the hint distribution $\mathcal{H}^{\text{sim}}$ as follows.

Hint distribution $\mathcal{H}_E^{\text{sim}}$

1. $\text{chl}, \varphi_1 \leftarrow D_{\text{chl}}(E)$. Let E_1 be the codomain of φ_1 .
2. $\psi, I \leftarrow \text{FastCommit}_{2^{e_1}, 3^{e_2}}$. Let E_2 be the codomain of ψ .
3. $\varphi_2 \leftarrow D_{\text{RspIsog}}(E_1, E_2)$, where $\varphi_2 : E_1 \rightarrow E_2$ or $\varphi_2 = \perp$.
4. If $\varphi_2 = \perp$, go to step 1.
5. Otherwise, write $\deg \varphi_2 = 2^n d'$ with d' odd.
6. $\varphi_3 \leftarrow D_{\text{AuxIsog}}(E_2, \psi, 2^{e_1-n} - d')$.
7. **return** $(\text{chl}, \varphi_2, \varphi_3)$.

We now prove that Σ_{Push} is $\mathcal{H}^{\text{sim}}$ -assisted wHVZK $^\perp$.

Theorem 3 Σ_{Push} is $\mathcal{H}^{\text{sim}}$ -assisted wHVZK $^\perp$.

Proof. We define a simulator $\mathcal{S}(E_{\text{pk}}, h) \rightarrow (E_{\text{com}}, \text{chl}, \text{rsp})$ as follows.

Input: A public key E_{pk} and a hint $h = (\text{chl}, \widehat{\varphi}_{\text{rsp}}, \varphi_{\text{aux}}) \leftarrow \mathcal{H}_{E_{\text{pk}}}^{\text{sim}}$.

Output: A transcript $(E_{\text{com}}, \text{chl}, \text{rsp})$.

1. Let E_{com} be the domain of φ_{rsp} and $\varphi_{\text{aux}}, E_{\text{chl}}$ be the codomain of φ_{rsp} , and E_{aux} be the codomain of φ_{aux} .
2. Compute $\deg \varphi_{\text{rsp}}$. Since $\deg \varphi_{\text{rsp}} < 2^{e_1}$, it is determined by the 2^{e_1} -Weil pairing on E_{com} and E_{chl} .
3. Write $\deg \varphi_{\text{rsp}} = 2^{n_1} d'_{\text{rsp}}$, where d'_{rsp} odd and let $d_{\text{aux}} = 2^{e_1-n_1} - d'_{\text{rsp}}$.
4. Take the deterministic basis $P_{\text{aux}}, Q_{\text{aux}}$ of $E_{\text{aux}}[2^{e_1}]$.
5. Let $P_{\text{chl}}, Q_{\text{chl}} \leftarrow [d_{\text{aux}}^{-1}] \varphi_{\text{rsp}} \circ \widehat{\varphi}_{\text{aux}}(P_{\text{aux}}), [d_{\text{aux}}^{-1}] \varphi_{\text{rsp}} \circ \widehat{\varphi}_{\text{aux}}(Q_{\text{aux}})$.
6. **return** $(E_{\text{com}}, \text{chl}, \text{rsp} = (E_{\text{aux}}, P_{\text{chl}}, Q_{\text{chl}}, n_1))$.

By construction, the distribution of the transcript $(E_{\text{com}}, \text{chl}, \text{rsp})$ output by the simulator $\mathcal{S}$ is exactly the same as that of the real transcript output by Σ_{Push} . $\square$

5.4 More Reasonable Hint Distribution

In order to prove the EUF-CMA security of our protocol, we need to assume that $(\mathcal{R}_{\text{OneEnd}}, \mathcal{H}^{\text{sim}})$ is a hard relation with hints (Definition 2). That is equal to the hardness of the following problem.

Problem 1 (q - $\mathcal{H}^{\text{sim}}$ -hint-OneEnd) *Given a supersingular elliptic curve E over $\mathbb{F}_{p^2}$ and q hints $h_1, \dots, h_q \leftarrow \mathcal{H}_E^{\text{sim}}$, find an endomorphism $\alpha \in \text{End}(E) \setminus \mathbb{Z}$.*

However, the hint distribution $\mathcal{H}^{\text{sim}}$ we used is too powerful; it solves an isogeny problem in D_{RspIsog} . Thus, we introduce another hint distribution $\mathcal{H}^{\text{unif}}$; rather than sampling a random curve and generating a connecting isogeny, the new hint distribution $\mathcal{H}_E^{\text{unif}}$ samples random isogenies directly from E . We show the hint distribution $\mathcal{H}^{\text{unif}}$ below.

Hint distribution $\mathcal{H}^{\text{unif}}$

1. Sample an integer d from a weighted distribution on $(0, 2^{e_1})_3$, where each integer d , with prime factorization $\prod_{i=1}^t p_i^{m_i}$, has weight $\prod_{i=1}^t (p_i + 1)^{m_i}$.
2. Sample an isogeny $\psi'_1 : E \rightarrow E_1$ uniformly among the (possibly non-cyclic) isogenies from E of degree d .
3. Write ψ_1 for the cyclic component of ψ'_1 .
4. Write $\deg \psi'_1 = 2^{n_1} \cdot d_1$, where d_1 is odd.
5. Write $2^{e_1 - n_1} - d_1 = 3^{n_2} \cdot d_2$, where d_2 is coprime to 3.
6. Sample an isogeny $\psi_2 : E_1 \rightarrow E_2$ uniformly among the isogenies from E_1 of degree d_2 .
7. **return** (ψ_1, ψ_2) .

Using a hint $h = (\psi_1, \psi_2)$ sampled from $\mathcal{H}_E^{\text{unif}}$, we can make something that looks like a hint from $\mathcal{H}_E^{\text{sim}}$ in polynomial time as follows:

1. Let E_1 be the codomain of ψ_1 , and let E_2 be the codomain of ψ_2 .
2. Let $(\text{chl}, \varphi) \leftarrow D_{\text{chl}}(E)$.
3. Write $\psi = \psi_2 \circ \psi_1 : E \rightarrow E_2$.
4. Compute the push-forward $\psi' : E' \rightarrow E'_2$ of ψ by φ . Since we can compute a generator of $\ker \varphi$ from chl , we can compute the push-forward as in Algorithm 2.
5. Decompose ψ' into $\psi' = \psi'_2 \circ \psi'_1$, where $\deg \psi'_i = \deg \psi_i$ for $i \in \{1, 2\}$. We can compute ψ'_1 and ψ'_2 using **KaniEval**.
6. Compute $d := \deg \psi'_1$. Since $\deg \psi'_1 < 2^{e_1}$, it is determined by the 2^{e_1} -Weil pairing on E and E_1 .
7. Write $d = 2^{n_1} \cdot d_1$, where d_1 odd.
8. Write $2^{e_1 - n_1} - d_1 = 3^{n_2} \cdot d_2$, where d_2 coprime to 3.
9. If n_2 is odd, let $\psi''_2 \leftarrow [3^{(n_2-1)/2}] \circ \varphi$ for a randomly chosen 3-isogeny φ uniformly among 3-isogenies from E'_2 . Otherwise, let $\psi''_2 \leftarrow [3^{n_2/2}]$.
10. **return** $(\text{chl}, \psi'_1, \psi''_2 \circ \psi'_2)$.

We denote this algorithm by **MakeHint**(h). Now, we define the problem to distinguish hints sampled from $\mathcal{H}^{\text{sim}}$ and the outputs of **MakeHint**.

Problem 2 (q -hint-dist) *Given a supersingular elliptic curve E over $\mathbb{F}_{p^2}$ and q hints $h_1, \dots, h_q$ sampled with probability $1/2$ from either distribution:*

- $\mathcal{D}_0 = \{(h_1, \dots, h_q) \mid h_i \leftarrow \mathcal{H}_E^{\text{sim}} \text{ for } i \in \{1, \dots, q\}\}.$
- $\mathcal{D}_1 = \{(h_1, \dots, h_q) \mid h'_i \leftarrow \mathcal{H}_E^{\text{unif}}, h_i \leftarrow \text{MakeHint}(h'_i) \text{ for } i \in \{1, \dots, q\}\}.$

Determine from which distribution the hints are sampled.

Under the hardness of Problem 2, the hardness of Problem 1 is equal to the hardness of the following problem.

Problem 3 (q - $\mathcal{H}^{\text{unif}}$ -hint-OneEnd) *Given a supersingular elliptic curve E over $\mathbb{F}_{p^2}$ and q hints $h_1, \dots, h_q \leftarrow \mathcal{H}_E^{\text{unif}}$, find an endomorphism $\alpha \in \text{End}(E) \setminus \mathbb{Z}$.*

Therefore, the hardness of Problem 2 and 3 implies the hardness of Problem 1. Finally, we have the following theorem.

Theorem 4 *Under Heuristic 1, Heuristic 2, and the hardness assumptions of Problem 2 and 3, SQIsign2DPush is EUF-CMA secure.*

5.5 Hardness Analysis

We discuss the hardness of Problem 2 and 3. We denote by $\mathcal{E}$ the set of all supersingular elliptic curves over $\mathbb{F}_{p^2}$ up to isomorphism. We also denote the uniform distribution over $\mathcal{E}$ by $\mathcal{E}^{\text{unif}}$.

Hardness of Problem 2: Let $h = (\text{chl}, \varphi_2, \varphi_3)$ be one of the q hints sampled from either $\mathcal{D}_0$ or $\mathcal{D}_1$. Let E_1 be the domain of φ_2 , let E_2 be the codomain of φ_2 (equal to the domain of φ_3), and let E_3 be the codomain of φ_3 . In addition, we let φ_1 be the isogeny with kernel $\langle P + [\text{chl}]Q \rangle$ for the deterministic basis (P, Q) of $E[3^{e_2}]$. Then φ_1 is the isogeny from E to E_1 .

First, we consider the case where $\mathcal{D}_{\text{RspIsog}}$ does not return $\perp$ in $\mathcal{H}_E^{\text{sim}}$. Then the distribution of φ_1 is exactly the same in both $\mathcal{D}_0$ and $\mathcal{D}_1$. Next, the conditional distribution of φ_2 given E_1 and E_2 is also the same in both cases: uniform among isogenies $E_1 \rightarrow E_2$ of degree $\in (0, 2^{e_1})_3$. Therefore, the only distinguishing factor for φ_2 is the distribution of E_2 . In $\mathcal{D}_0$, the curve E_2 is the output of $\text{FastCommit}_{2^{e_1}, 3^{e_2}}$ and its distribution is considered close to $\mathcal{E}^{\text{unif}}$ from Heuristic 1. In $\mathcal{D}_1$, the curve E_2 is the codomain of a random isogeny φ_2 from E_1 of bounded degree and its distribution is also considered close to $\mathcal{E}^{\text{unif}}$ as discussed in [2, Remark 5.1]. Now, the only distinguishing factor is φ_3 . Write $\deg \varphi_3 = 3^n d'$, where d' coprime to 3 and decompose φ_3 into $\varphi_3^{(0)} \circ \varphi_3^{(1)}$ such that $\deg \varphi_3^{(0)} = 3^n$ and $\deg \varphi_3^{(1)} = d'$. Since the 3^n -isogeny $\varphi_3^{(0)}$ is sampled in the same way in both cases, the only difference is in the distribution of $\varphi_3^{(1)}$: in $\mathcal{D}_0$ is output of PushRandIsog , while in $\mathcal{D}_1$ is distributed uniformly among d' -isogenies from E_2 . Recall that PushRandIsog computes the push-forward of a d' -isogeny φ_0 by an isogeny $\psi : E_0 \rightarrow E_2$, where φ_0 is computed by a similar way as RandIsogImg , the QFESTA's algorithm [31, Algorithm 2]. By a similar

discussion for the output distribution of `RandIsogImg` [31, Section 3.1], the distribution of φ_0 is considered close to the uniform distribution among d' -isogenies from E_0 . Therefore, the output distribution of `PushRandIsog` is also considered close to uniform.

Next, we discuss the effect of the failure in $\mathcal{D}_{\text{RspIsog}}$ on the distribution of (E_1, E_2) . As in the previous paragraph, we assume Heuristic 1. We first discuss the distribution of E_1 . Let $\mathcal{C}_E$ be the set of possible curves E_1 computed by $D_{\text{chl}}(E)$. Additionally, for a curve $E_1 \in \mathcal{C}_E$, we denote by $\mathcal{E}(E_1)$ the set of curves $E_2 \in \mathcal{E}$ such that there exists an isogeny $E_1 \rightarrow E_2$ of degree $\in (0, 2^{e_1})_3$. Then (E_1, E_2) in $\mathcal{D}_0$ is distributed uniformly over $\mathcal{P}_E = \{(E_1, E_2) \mid E_1 \in \mathcal{C}_E, E_2 \in \mathcal{E}(E_1)\} \subseteq \mathcal{C}_E \times \mathcal{E}$. Therefore, the distribution of E_1 in $\mathcal{D}_0$ is weighted as follows:

$$\Pr[E_1 = E'_1] = \frac{\#\mathcal{E}(E'_1)}{\#\mathcal{P}_E} \text{ for } E'_1 \in \mathcal{C}_E.$$

We denote this distribution by $\mathcal{C}_E^{\text{sim}}$. On the other hand, the distribution of E_1 computed in $\mathcal{D}_1$ is uniform over $\mathcal{C}_E$. We denote the uniform distribution over $\mathcal{C}_E$ by $\mathcal{C}_E^{\text{unif}}$. Then the statistical distance between $\mathcal{C}_E^{\text{sim}}$ and $\mathcal{C}_E^{\text{unif}}$ is

$$\begin{aligned} \Delta(\mathcal{C}_E^{\text{sim}}, \mathcal{C}_E^{\text{unif}}) &= \frac{1}{2} \sum_{E_1 \in \mathcal{C}_E} \left| \frac{\#\mathcal{E}(E_1)}{\#\mathcal{P}_E} - \frac{1}{\#\mathcal{C}_E} \right| \\ &\leq \frac{1}{2} \#\mathcal{C}_E \max_{E_1, E'_1 \in \mathcal{C}_E} \left| \frac{\#\mathcal{E}(E_1)}{\#\mathcal{P}_E} - \frac{\#\mathcal{E}(E'_1)}{\#\mathcal{P}_E} \right| \\ &= \frac{1}{2} \frac{\#(\mathcal{C}_E \times \mathcal{E})}{\#\mathcal{P}_E} \frac{1}{\#\mathcal{E}} \max_{E_1, E'_1 \in \mathcal{C}_E} |\#\mathcal{E}(E_1) - \#\mathcal{E}(E'_1)| \\ &= \frac{\max_{E_1, E'_1 \in \mathcal{C}_E} |\#\mathcal{E}(E_1) - \#\mathcal{E}(E'_1)|}{2 \Pr[\perp \leftarrow D_{\text{RspIsog}}]} \cdot \frac{1}{\#\mathcal{E}}, \end{aligned}$$

where we used $\Pr[\perp \leftarrow D_{\text{RspIsog}}] = \frac{\#\mathcal{P}_E}{\#(\mathcal{C}_E \times \mathcal{E})}$. From Heuristic 2, we have $\Pr[\perp \leftarrow D_{\text{RspIsog}}] \geq 2/3$. In addition, we have $\#\mathcal{E} \approx 2^{2\lambda}$. Therefore, if for any $E_1, E'_1 \in \mathcal{C}_E$, $|\#\mathcal{E}(E_1) - \#\mathcal{E}(E'_1)|$ is bounded by $o(2^\lambda)$, the distance $\Delta(\mathcal{C}_E^{\text{sim}}, \mathcal{C}_E^{\text{unif}})$ is negligible. We believe that $|\#\mathcal{E}(E_1) - \#\mathcal{E}(E'_1)|$ does not become so large, but the detailed analysis is a future work.

Finally, we discuss the conditional distribution of E_2 given $E_1 \in \mathcal{C}_E$. In $\mathcal{D}_0$, the conditional distribution of E_2 is uniform over $\mathcal{E}(E_1)$. On the other hand, in $\mathcal{D}_1$, E_2 is the codomain of a random isogeny φ_2 from E_1 of degree $\in (0, 2^{e_1})_3$ and its distribution is considered to be close to the uniform distribution over $\mathcal{E}(E_1)$.

Hardness of Problem 3: Our definition of $\mathcal{H}^{\text{unif}}$ is similar to that of `SQIsign-v2.0` [1, Algorithm 5]: the essential difference is that we sample an integer d to be coprime to 3. We do not expect this difference provides any help in finding an endomorphism. Therefore, q - $\mathcal{H}^{\text{unif}}$ -hint-OneEnd is considered to be as hard as OneEnd (without hints) from the same discussion in [2, Section 5.5]. OneEnd is believed to computationally hard from [38, 33]

5.6 λ -Bit Security

We ensure that our protocol achieves λ -bit security under our parameter settings. From the previous discussion, $q\text{-}\mathcal{H}^{\text{sim}}\text{-hint-OneEnd}$ is considered to be as hard as OneEnd. The best known algorithm to solve OneEnd requires $\tilde{O}(\sqrt{p})$ computations [21]. Since $p \approx 2^{2\lambda}$ in our setting, solving OneEnd requires $\tilde{O}(2^\lambda)$ computations. In this attack, the most computationally intensive task involves searching for a path in the supersingular isogeny graph to a curve within a specific set, which has an approximate cardinality of $\tilde{O}(p^{1/2})$. Therefore, a quantum adversary has the potential to reduce the computational costs of these attacks to $\tilde{O}(p^{1/4}) = \tilde{O}(2^{\lambda/2})$ using the Grover's algorithm.

Next, we consider the attack to find an isogeny between E_0 and E_{pk} (or E_{com}). We can use the Delfs-Galbraith method [17] to find an isogeny between these two curves. Its cost is $\tilde{O}(p^{1/2}) = \tilde{O}(2^\lambda)$. In the case of a quantum adversary, the cost can be reduced to $\tilde{O}(p^{1/4}) = \tilde{O}(2^{\lambda/2})$ by applying the Grover's algorithm. In addition, we know that there exists a smooth degree (2^{2e_1} - or 3^{2e_2} -) isogeny. Therefore, using the meet-in-the-middle attack [23], one can find an isogeny with a cost of $\tilde{O}(2^{e_1})$ or $\tilde{O}(3^{e_2})$. Since $2^{e_1} \approx 3^{e_2} \approx 2^\lambda$ in our setting, this attack also requires $\tilde{O}(2^\lambda)$ computations. In the quantum setting, by using the method in [35], the cost is reduced to $\tilde{O}(2^{2\lambda/3})$.

Finally, we consider the attack against the hash function that outputs a challenge. Our challenge space is $[0, 3^{e_2})$, and its size is approximately $3^{e_2} \approx 2^\lambda$, which is sufficient to achieve λ -bit security. However, the actual parameters shown in Section 6.1 indicate that $\log_2 3^{e_2}$ is slightly smaller than λ . Therefore, we apply the *grinding technique*: to compensate for the $\lambda - \log_2 3^{e_2}$ missing bits, the hash function is constructed by iterating $\lceil 2^\lambda / 3^{e_2} \rceil$ times a standard hash function. Note that the grinding technique is also used in SQIsign-v2.0. In the case of a quantum adversary, the Grover's algorithm reduces the cost to $\tilde{O}(2^{\lambda/2})$.

6 Efficiency

In this section, we analyze the efficiency of SQIsign2DPush. First, we provide concrete parameters of our protocol for NIST security level 1, 3, and 5. Then we compare the data sizes of our protocol with SQIsign-v2.0, SQIsign2D-East, and SQIsign2D². Finally, we compare the computational cost of SQIsign2DPush. Note that we consider the compressed versions for SQIsign2D-East and SQIsign2D².

6.1 Parameters

We give concrete parameters of SQIsign2DPush satisfying the NIST security level 1, 3, and 5. We also show the number of iterations of the hash function for the grinding technique.

Security	e_1	e_2	c	p	Hash iterations
Level 1	131	78	1	$2^{131} \cdot 3^{78} - 1$	21
Level 3	191	117	1	$2^{191} \cdot 3^{117} - 1$	95
Level 5	263	156	1	$2^{263} \cdot 3^{156} - 1$	430

Table 2. Public key and signature size comparison in bytes

Security	Protocol	Public key	Signature
Level 1	SQIsign-v2.0	65	148
	SQIsign2D-East	66	164
	SQIsign2D ²	64	154
	SQIsign2DPush	66	151
Level 3	SQIsign-v2.0	97	224
	SQIsign2D-East	98	245
	SQIsign2D ²	98	228
	SQIsign2DPush	98	219
Level 5	SQIsign-v2.0	129	292
	SQIsign2D-East	128	331
	SQIsign2D ²	128	299
	SQIsign2DPush	130	297

6.2 Data Sizes

In this subsection, we compare the data sizes of SQIsign-v2.0, SQIsign2D-East, SQIsign2D², and our SQIsign2DPush using the above parameters. Table 2 shows the sizes of each public key and signature. As shown in Table 2, the signature size of SQIsign2DPush is comparable to the other protocols. The public key size of our protocol is almost the same as the other protocols.

6.3 Computational Cost

We compare the computational costs in security level 1. Table 3 shows the number of isogeny computations of each degree. In this comparison, we also take into account SQIsign-v2.0 with Qlapoti [8].

As for the signing cost, our protocol requires the smallest number of $(2, 2)$ -isogeny computations among the signature schemes in the table. In particular, compared to SQIsign-v2.0 and SQIsign2D-East, it is less than $1/4$ and $1/3$, respectively. Though our protocol requires relatively large number of 1-dimensional isogeny computations, these costs are expected to be smaller than that of $(2, 2)$ -isogeny computations. Therefore, the total signing cost of our protocol expected to be smaller than SQIsign-v2.0 and SQIsign2D-East. On the other hand, when compared to SQIsign-v2.0 with Qlapoti and SQIsign2D², the reduction in the number of $(2, 2)$ -isogeny computations is limited to $1/2$ and $2/3$, respectively. In this case, we cannot ignore the impact of the 1-dimensional isogeny, making it difficult to determine a clear advantage. However, the length of the isogeny chain computed in **FastCommit** of our protocol can be further reduced as long as it remains resistant to the meet-in-the-middle attack. This optimization could

Table 3. Number of isogeny computations of each degree. The numbers in parentheses mean that they may vary slightly depending on the case.

Protocol (Security level 1)		2	3	(2, 2)
SQIsign-v2.0	Gen	-	-	500
	Sign	(248)	-	1126
	Verf	(248)	-	(126)
SQIsign-v2.0 + Qlapoti [8]	Gen	-	-	246
	Sign	(248)	-	622
	Verf	(248)	-	(126)
SQIsign2D-East	Gen	-	-	253
	Sign	126	(2)	760
	Verf	126	(2)	127
SQIsign2D ²	Gen	65	78	66
	Sign	-	(390)	393
	Verf	-	78	131
SQIsign2DPush	Gen	393	234	-
	Sign	(262)	468	(262)
	Verf	(0)	78	(131)

allow our scheme to outperform these two schemes. A detailed comparison with these schemes is left as future work.

As for the verification, the number of $(2, 2)$ -isogeny computations of our protocol is comparable to the others. However, we need to compute one point image during the $(2, 2)$ -isogeny computations in our protocol. Therefore, our computational cost is considered to be slightly higher.

Finally, we developed a proof-of-concept implementation of SQIsign2DPush in Julia language [7] with its computer algebra package Nemo [22]. The implementation is available at:

<https://github.com/hiroshi-onuki/SQIsign2D-Push>.

Table 4 shows the computational times of the implementation. These are the averages of 100 run times. The computational times are measured on a computer with an Intel Core i7-10700K CPU@3.70Hz without Turbo Boost. The cost evaluation through an optimized implementation is a future work.

Table 4. Computational times (sec.)

Security	Gen	Sign	Verf
Level 1	0.28	0.85	0.14
Level 3	0.50	1.62	0.33
Level 5	1.04	2.60	0.45

7 Conclusion

In this paper, we introduce SQIsign2DPush, a new variant of SQIsign with faster signing. As a building block of our protocol, we construct a new algorithm named **PushRandIsog**, which computes a given (generally non-smooth)

degree isogeny from a given elliptic curve. Using `PushRandIsog`, we compute an auxiliary isogeny faster than the other variants of `SQIsign` and reduce the computational cost of signing.

Indeed, the number of $(2, 2)$ -isogeny computations for signing of our protocol is less than $1/4$ of `SQIsign-v2.0`. Although `SQIsign-v2.0` could potentially outperform our protocol when utilizing `Qlapoti`, the optimization of `FastCommit` may enable our scheme to achieve superior performance. The verification cost and the signature size of our protocol is comparable to `SQIsign-v2.0`.

As a future work, we need to evaluate the failure probability of the response in more detail. The cost evaluation and comparison through an optimized implementation is also a future work.

Acknowledgements

This work was supported by JST K Program Grant Number JPMJKP24U2, Japan.

References

1. Marius A. Aardal, Gora Adj, Diego F. Aranha, Andrea Basso, Isaac Andrés Canales Martínez, Jorge Chávez-Saab, Maria Corte-Real Santos, Pierrick Dartois, Luca De Feo, Max Duparc, Jonathan Komada Eriksen, Tako Boris Fouotsa, Décio Luiz Gazzoni Filho, Basil Hess, David Kohel, Antonin Leroux, Patrick Longa, Luciano Maino, Michael Meyer, Kohei Nakagawa, Hiroshi Onuki, Lorenz Panny, Sikhar Patranabis, Christophe Petit, Giacomo Pope, Krijn Reijnders, Damien Robert, Francisco Rodríguez Henríquez, Sina Schaeffler, and Benjamin Wesolowski. `SQIsign`. Technical report, National Institute of Standards and Technology, 2024. available at <https://csrc.nist.gov/Projects/pqc-dig-sig/round-2-additional-signatures>.
2. Marius A. Aardal, Andrea Basso, Luca De Feo, Sikhar Patranabis, and Benjamin Wesolowski. A complete security proof of `SQIsign`. *Cryptology ePrint Archive*, Report 2025/379, 2025.
3. Michel Abdalla, Jee Hea An, Mihir Bellare, and Chanathip Namprempre. From identification to signatures via the Fiat-Shamir transform: Minimizing assumptions for security and forward-security. In Lars R. Knudsen, editor, *EUROCRYPT 2002*, volume 2332 of *LNCS*, pages 418–433. Springer, Berlin, Heidelberg, April / May 2002.
4. Andrea Basso, Pierrick Dartois, Luca De Feo, Antonin Leroux, Luciano Maino, Giacomo Pope, Damien Robert, and Benjamin Wesolowski. `SQIsign2D-West` - the fast, the small, and the safer. In Kai-Min Chung and Yu Sasaki, editors, *ASIACRYPT 2024, Part III*, volume 15486 of *LNCS*, pages 339–370. Springer, Singapore, December 2024.
5. Andrea Basso and Luciano Maino. `POKÉ`: A compact and efficient pke from higher-dimensional isogenies. In Serge Fehr and Pierre-Alain Fouque, editors, *EUROCRYPT 2025*, volume 15602 of *LNCS*, pages 94–123. Springer, Cham, May 2025.

6. Andrea Basso, Luciano Maino, and Giacomo Pope. FESTA: Fast encryption from supersingular torsion attacks. In Jian Guo and Ron Steinfeld, editors, *ASIACRYPT 2023, Part VII*, volume 14444 of *LNCS*, pages 98–126. Springer, Singapore, December 2023.
7. Jeff Bezanson, Alan Edelman, Stefan Karpinski, and Viral B Shah. Julia: A fresh approach to numerical computing. *SIAM Review*, 59(1):65–98, 2017.
8. Giacomo Borin, Maria Corte-Real Santos, Jonathan Komada Eriksen, Riccardo Invernizzi, Marzio Mula, Sina Schaeffler, and Frederik Vercauteren. Qlapoti: Simple and Efficient Translation of Quaternion Ideals to Isogenies. *Cryptology ePrint Archive, Report 2025/1604*, 2025.
9. Wouter Castryck and Thomas Decru. An efficient key recovery attack on SIDH. In Carmit Hazay and Martijn Stam, editors, *EUROCRYPT 2023, Part V*, volume 14008 of *LNCS*, pages 423–447. Springer, Cham, April 2023.
10. Jorge Chavez-Saab, Maria Corte-Real Santos, Luca De Feo, Jonathan Komada Eriksen, Basil Hess, David Kohel, Antonin Leroux, Patrick Longa, Michael Meyer, Lorenz Panny, Sikhar Patranabis, Christophe Petit, Francisco Rodríguez Henríquez, Sina Schaeffler, and Benjamin Wesolowski. SQISign. Technical report, National Institute of Standards and Technology, 2023. available at <https://csrc.nist.gov/Projects/pqc-dig-sig/round-1-additional-signatures>.
11. Craig Costello and Hüseyin Hisil. A simple and compact algorithm for SIDH with arbitrary degree isogenies. In Tsuyoshi Takagi and Thomas Peyrin, editors, *ASIACRYPT 2017, Part II*, volume 10625 of *LNCS*, pages 303–329. Springer, Cham, December 2017.
12. Craig Costello, David Jao, Patrick Longa, Michael Naehrig, Joost Renes, and David Urbanik. Efficient compression of SIDH public keys. In Jean-Sébastien Coron and Jesper Buus Nielsen, editors, *EUROCRYPT 2017, Part I*, volume 10210 of *LNCS*, pages 679–706. Springer, Cham, April / May 2017.
13. Pierrick Dartois, Antonin Leroux, Damien Robert, and Benjamin Wesolowski. SQISignHD: New dimensions in cryptography. In Marc Joye and Gregor Leander, editors, *EUROCRYPT 2024, Part I*, volume 14651 of *LNCS*, pages 3–32. Springer, Cham, May 2024.
14. Pierrick Dartois, Luciano Maino, Giacomo Pope, and Damien Robert. An algorithmic approach to $(2, 2)$ -isogenies in the theta model and applications to isogeny-based cryptography. In Kai-Min Chung and Yu Sasaki, editors, *ASIACRYPT 2024, Part III*, volume 15486 of *LNCS*, pages 304–338. Springer, Singapore, December 2024.
15. Luca De Feo, David Kohel, Antonin Leroux, Christophe Petit, and Benjamin Wesolowski. SQISign: Compact post-quantum signatures from quaternions and isogenies. In Shiho Moriai and Huaxiong Wang, editors, *ASIACRYPT 2020, Part I*, volume 12491 of *LNCS*, pages 64–93. Springer, Cham, December 2020.
16. Luca De Feo, Antonin Leroux, Patrick Longa, and Benjamin Wesolowski. New algorithms for the Deuring correspondence - towards practical and secure SQISign signatures. In Carmit Hazay and Martijn Stam, editors, *EUROCRYPT 2023, Part V*, volume 14008 of *LNCS*, pages 659–690. Springer, Cham, April 2023.
17. Christina Delfs and Steven D. Galbraith. Computing isogenies between supersingular elliptic curves over $\mathbb{F}_p$. *DCC*, 78(2):425–440, 2016.
18. Max Deuring. Die typen der multiplikatorenringe elliptischer funktionenkörper. *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg*, 14:197–272, 1941.

19. Max Duparc and Tako Boris Fouotsa. SQIPrime: A dimension 2 variant of SQISignHD with non-smooth challenge isogenies. In Kai-Min Chung and Yu Sasaki, editors, *ASIACRYPT 2024, Part III*, volume 15486 of *LNCS*, pages 396–429. Springer, Singapore, December 2024.
20. Pierre Dusart. The k th prime is greater than $k(\ln k + \ln \ln k - 1)$ for $k \geq 2$. *Mathematics of Computation*, 68(225):411–415, 1999.
21. Kirsten Eisenträger, Sean Hallgren, Chris Leonardi, Travis Morrison, and Jennifer Park. Computing endomorphism rings of supersingular elliptic curves and connections to path-finding in isogeny graphs. *Open Book Series*, 4(1):215–232, 2020.
22. Claus Fieker, William Hart, Tommy Hofmann, and Fredrik Johansson. Nemo/hecke: Computer algebra and number theory packages for the julia programming language. In *Proceedings of the 2017 ACM on International Symposium on Symbolic and Algebraic Computation*, ISSAC '17, pages 157–164, New York, NY, USA, 2017. ACM.
23. Steven D Galbraith. Constructing isogenies between elliptic curves over finite fields. *LMS Journal of Computation and Mathematics*, 2:118–138, 1999.
24. Wissam Ghantous. Loops, multi-edges and collisions in supersingular isogeny graphs, 2024.
25. David Jao, Reza Azarderakhsh, Matthew Campagna, Craig Costello, Luca De Feo, Basil Hess, Amir Jalali, Brian Koziel, Brian LaMacchia, Patrick Longa, Michael Naehrig, Joost Renes, Vladimir Soukharev, David Urbanik, Geovandro Pereira, Koray Karabina, and Aaron Hutchinson. SIKE. Technical report, National Institute of Standards and Technology, 2020. available at <https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization/round-3-submissions>.
26. David Jao and Luca De Feo. Towards quantum-resistant cryptosystems from supersingular elliptic curve isogenies. In Bo-Yin Yang, editor, *Post-Quantum Cryptography - 4th International Workshop, PQCrypto 2011*, pages 19–34. Springer, Berlin, Heidelberg, November / December 2011.
27. Ernst Kani. The number of curves of genus two with elliptic differentials. *Journal für die reine und angewandte Mathematik*, 1997(485):93–122, 1997.
28. David Kohel, Kristin Lauter, Christophe Petit, and Jean-Pierre Tignol. On the quaternion-isogeny path problem. *LMS Journal of Computation and Mathematics*, 17(A):418–432, 2014.
29. Vadim Lyubashevsky. Fiat-Shamir with aborts: Applications to lattice and factoring-based signatures. In Mitsuru Matsui, editor, *ASIACRYPT 2009*, volume 5912 of *LNCS*, pages 598–616. Springer, Berlin, Heidelberg, December 2009.
30. Luciano Maino, Chloe Martindale, Lorenz Panny, Giacomo Pope, and Benjamin Wesolowski. A direct key recovery attack on SIDH. In Carmit Hazay and Martijn Stam, editors, *EUROCRYPT 2023, Part V*, volume 14008 of *LNCS*, pages 448–471. Springer, Cham, April 2023.
31. Kohei Nakagawa and Hiroshi Onuki. QFESTA: Efficient algorithms and parameters for FESTA using quaternion algebras. In Leonid Reyzin and Douglas Stebila, editors, *CRYPTO 2024, Part V*, volume 14924 of *LNCS*, pages 75–106. Springer, Cham, August 2024.
32. Kohei Nakagawa, Hiroshi Onuki, Wouter Castryck, Mingjie Chen, Riccardo Invernizzi, Gioella Lorenzon, and Frederik Vercauteren. SQISign2D-East: A new signature scheme using 2-dimensional isogenies. In Kai-Min Chung and Yu Sasaki, editors, *ASIACRYPT 2024, Part III*, volume 15486 of *LNCS*, pages 272–303. Springer, Singapore, December 2024.

33. Aurel Page and Benjamin Wesolowski. The supersingular endomorphism ring and one endomorphism problems are equivalent. In Marc Joye and Gregor Leander, editors, *EUROCRYPT 2024, Part VI*, volume 14656 of *LNCS*, pages 388–417. Springer, Cham, May 2024.
34. Damien Robert. Breaking SIDH in polynomial time. In Carmit Hazay and Martijn Stam, editors, *EUROCRYPT 2023, Part V*, volume 14008 of *LNCS*, pages 472–503. Springer, Cham, April 2023.
35. Seiichiro Tani. Claw finding algorithms using quantum walk. *Theoretical Computer Science*, 410(50):5285–5297, 2009.
36. Enrique Treviño. The least inert prime in a real quadratic field. *Mathematics of Computation*, 81(279):1777–1797, 2012.
37. Jacques Vélú. Isogénies entre courbes elliptiques. *Comptes-Rendus de l’Académie des Sciences*, 273:238–241, 1971.
38. Benjamin Wesolowski. The supersingular isogeny path and endomorphism ring problems are equivalent. In *62nd FOCS*, pages 1100–1111. IEEE Computer Society Press, February 2022.
39. Zheng Xu, Kaizhan Lin, Chang-An Zhao, and Yi Ouyang. SQIsign2D²: New SQIsign2D Variant by Leveraging Power Smooth Isogenies in Dimension One. *Cryptology ePrint Archive, Report 2025/920*, 2025.

A Fundamental Cryptographic Definitions

In this section, we give definitions of Σ -protocol, signature scheme, and its EUF-CMA security.

Let a relation $\mathcal{R}$ be a set of tuples $(x, w) \in \mathcal{X} \times \mathcal{W} \subseteq \{0, 1\}^* \times \{0, 1\}^*$ satisfying the following conditions:

- There exists a PPT algorithm which takes $(x, w) \in \{0, 1\}^* \times \{0, 1\}^*$ as input and determines whether $(x, w) \in \mathcal{R}$ or not.
- There exists a PPT algorithm $\mathbf{Gen}_{\mathcal{R}}$ called an instance generator which takes 1^λ as input and returns $(x, w) \in \mathcal{R}$ such that $|x| = \text{poly}(\lambda)$.

Definition 6 (Σ -protocol). A Σ -protocol for a relation $\mathcal{R}$ is a 2-round interactive protocol between two parties: a prover and a verifier, given by a tuple of three PPT algorithms $\Sigma = (P_1, P_2, V)$ as follows.

1. Given $(x, y) \leftarrow \mathbf{Gen}_{\mathcal{R}}(1^\lambda)$, the prover runs $(\text{com}, \text{st}) \leftarrow P_1(x, w)$ and obtains a commitment com and a state st . The prover sends com to the verifier.
2. The verifier samples a challenge chl uniformly from a set $\mathcal{C}$ called the challenge space and sends chl to the prover.
3. The prover runs $\text{rsp} \leftarrow P_2(x, w, \text{st}, \text{chl})$ and sends rsp to the verifier.
4. Finally, the verifier runs $b \leftarrow V(x, \text{com}, \text{chl}, \text{rsp})$ and obtains a bit b . The verifier accepts if and only if $b = 1$.

In addition, the Σ -protocol should have correctness: When the prover and the verifier run the above procedure honestly, the verifier accepts except with negligible probability in λ .

Definition 7 (Signature scheme). A signature scheme is a tuple of three PPT algorithms $S = (\mathbf{Gen}, \mathbf{Sign}, \mathbf{Verf})$ as follows.

- $\mathbf{Gen}(1^\lambda) \rightarrow (\text{pk}, \text{sk})$: The key generation algorithm which takes a security parameter 1^λ as input and returns a public key pk and a secret key sk .
- $\mathbf{Sign}(\text{sk}, m) \rightarrow \text{sig}$: The signing algorithm which takes a secret key sk and a message m as input and returns a signature sig .
- $\mathbf{Verf}(\text{pk}, m, \text{sig})$: The verification algorithm which takes a public key pk , a message m , and a signature sig as input, and returns 1(accept) or 0(reject).

In addition, the signature scheme should have correctness: For a correctly generated signature, the verification algorithm outputs 1(accept) except with negligible probability in λ .

Furthermore, one of the important security properties that a signature scheme should possess is existential unforgeability under chosen message attack (EUF-CMA).

Definition 8 (EUF-CMA). Let $S = (\mathbf{Gen}, \mathbf{Sign}, \mathbf{Verf})$ be a signature scheme. We say that S is EUF-CMA secure if for all PPT algorithm $\mathcal{A}$, it holds that

$$\text{Adv}_S^{\text{EUF-CMA}}(\mathcal{A}) := \Pr[\text{EUF-CMA}_S(\mathcal{A}) = 1] = \text{negl}(\lambda),$$

where EUF-CMA_S is the following game.

Game: EUF-CMA_S

1. $(pk, sk) \leftarrow \mathbf{Gen}(1^\lambda)$
2. $\mathcal{Q} \leftarrow \emptyset$
3. $(m^*, \text{sig}^*) \leftarrow \mathcal{A}^{\text{OSign}}(pk)$
4. **if** $\mathbf{Verf}(pk, m, \text{sig}) = 1$ and $m^* \notin \mathcal{Q}$
 - 3-1. **return** 1
5. **else**
 - 4-1. **return** 0

OSign(m):

1. $\text{sig} \leftarrow \mathbf{Sign}(sk, m)$
2. $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{m\}$
3. **return** sig

B Omitted Proofs

In this section, we give omitted proofs for Lemma 1 and Lemma 2.

B.1 Proof for Lemma 1

Proof. The construction of $\mathcal{B}$ is similar to that of [3, Lemma 3.5]. We denote the output of $\mathcal{A}$ by $(m^*, \text{sig}^* = (\text{com}^*, \text{rsp}^*))$. Let $\mathcal{A}'$ be an algorithm that is almost the same as $\mathcal{A}$ but queries $\mathbf{R0}(pk||m^*||\text{com}^*)$ at least once. We can construct such $\mathcal{A}'$ by adding the query $\mathbf{R0}(pk||m^*||\text{com}^*)$ just before outputting (m^*, sig^*) . Note that the number of queries $\mathcal{A}'$ makes to $\mathbf{R0}$ and $\mathbf{OSign}$ are at most $q + 1$ and s , respectively.

$\mathcal{B} = (\mathcal{B}_1, \mathcal{B}_2)$ communicates with $\mathcal{A}'$ and operates as follows. $\mathcal{B}$ first samples $h \xleftarrow{\$} [1, q + 1]$ and pre-computes s transcripts $(\text{com}_i, \text{chl}_i, \text{rsp}_i)_{i \in [1, s]}$ using $\mathbf{OTrans}_{pk, sk}^\perp$. $\mathcal{B}$ answers $(q + 1)$ -times $\mathbf{R0}$ queries of $\mathcal{A}$ using a hash table HT . At the time of the h -th query with input $x = pk||m^\dagger||\text{com}^\dagger$, $\mathcal{B}$ returns $\text{com}^\dagger$ as the output of $\mathcal{B}_1$. Then $\mathcal{B}$ obtains $\text{chl} \in \mathcal{C}$ as an input of $\mathcal{B}_2$. $\mathcal{B}$ also answers s -times $\mathbf{OSign}$ queries of $\mathcal{A}$ using the pre-computed transcripts. At each query $\mathbf{OSign}(m_i)$, $\mathcal{B}$ checks whether $HT[pk||m_i||\text{com}_i] = \perp$. If so, $\mathcal{B}$ sets $HT[pk||m_i||\text{com}_i] = \text{chl}_i$ and answers $(\text{com}_i, \text{rsp}_i)$, otherwise it fails. Finally, $\mathcal{B}$ obtains $(m^*, (\text{com}^*, \text{rsp}^*))$ as the output of $\mathcal{A}'$. If $(m^\dagger, \text{com}^\dagger) = (m^*, \text{com}^*)$, $\mathcal{B}$ returns rsp^* as the output of $\mathcal{B}_2$, otherwise it fails.

Let $(\text{com}_{i,j}, \text{chl}_{i,j})$ be the pair of commitment and challenge chosen in the j -th while-loop of the i -th $\mathbf{OTrans}^\perp$ query. We also denote by B_i the number of iterations of the while-loop executed within the i -th $\mathbf{OTrans}^\perp$ query. If $\mathcal{A}'$ queries $\mathbf{R0}$ with input $pk||m_i||\text{com}_{i,j}$ for $j < B_i$, $\mathcal{B}$ should set $HT[pk||m_i||\text{com}_{i,j}] = \text{chl}_{i,j}$. However, $\mathcal{B}$ does not know $\text{chl}_{i,j}$ for $j < B_i$. Therefore, $\mathcal{B}$ cannot simulate $\mathbf{R0}$ with such input. We denote such event by $\text{bad}_{\mathbf{R0}}$. In addition, we denote by $\text{bad}_{\mathbf{OSign}}$ the event that $\mathcal{B}$ fails in answering a $\mathbf{OSign}$ query, which occurs when $HT[pk||m_i||\text{com}_i]$ is already set. Since the probability that $(m^\dagger, \text{com}^\dagger) = (m^*, \text{com}^*)$ holds is $\frac{1}{q+1}$, we have the following inequality:

$$\text{Adv}_{\Sigma}^{\text{IMP-PA}^\perp}(\mathcal{B}) \geq \frac{1}{q+1} \left(\text{Adv}_{\text{SIG}^\perp[\Sigma]}^{\text{EUF-CMA}}(\mathcal{A}) - \Pr[\text{bad}_{\mathbf{R0}} \vee \text{bad}_{\mathbf{OSign}}] \right).$$

In addition, we denote by $\mathbf{bad}_B$ the event that at least one of $\mathbf{OTrans}^\perp$ queried by $\mathcal{B}$ performs a while-loop more than B times internally. Then we have

$$\begin{aligned} \Pr[\mathbf{bad}_{R0} \vee \mathbf{bad}_{0\text{Sign}}] &= \Pr[(\mathbf{bad}_{R0} \vee \mathbf{bad}_{0\text{Sign}}) \wedge \neg \mathbf{bad}_B] \\ &\quad + \Pr[(\mathbf{bad}_{R0} \vee \mathbf{bad}_{0\text{Sign}}) \wedge \mathbf{bad}_B] \\ &\leq \Pr[\mathbf{bad}_{R0} \vee \mathbf{bad}_{0\text{Sign}} \mid \neg \mathbf{bad}_B] + \Pr[\mathbf{bad}_B] \\ &\leq \Pr[\mathbf{bad}_{R0} \vee \mathbf{bad}_{0\text{Sign}} \mid \neg \mathbf{bad}_B] + s \cdot (\mathcal{P}_\Sigma^\perp)^B. \end{aligned}$$

When $\neg \mathbf{bad}_B$, $\mathbf{OTrans}^\perp$ generates $\text{com}_{i,j}$ at most Bs times internally. If all of these $\text{com}_{i,j}$ are different from the commitments appeared in the past while-loop and those queried to $R0$, then $\mathbf{bad}_{R0} \vee \mathbf{bad}_{0\text{Sign}}$ cannot occur. The number of the commitments which appear before $\text{com}_{i,j}$, considering those queried to $R0$, is at most $q + 1 + B(i - 1) + j - 1 = q + B(i - 1) + j$. Therefore, we have

$$\begin{aligned} \Pr[\mathbf{bad}_{R0} \vee \mathbf{bad}_{0\text{Sign}} \mid \neg \mathbf{bad}_B] &\leq \sum_{i=1}^s \sum_{j=1}^B (q + B(i - 1) + j) \cdot \text{MinEnt}(\Sigma) \\ &\leq (1 + q + Bs)Bs \cdot \text{MinEnt}(\Sigma). \end{aligned}$$

□

B.2 Proof for Lemma 2

The proof for this lemma is almost the same as that of [2, Lemma 3.1]. Using $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ and $\mathcal{S}$, we construct a PPT two-stage algorithm $\mathcal{B} = (\mathcal{B}_1, \mathcal{B}_2)$ against hint-IMP-PA game as follows.

$\mathcal{B}_1$ first takes $(x, h_1, \dots, h_s)$ as inputs and sends x to $\mathcal{A}_1$. For the i -th query of $\mathbf{OTrans}^\perp$ by $\mathcal{A}_1$, $\mathcal{B}_1$ runs $\pi_i \leftarrow \mathcal{S}(x, h_i)$ and responds with π_i . When $\mathcal{A}_1$ terminates and outputs $(\text{com}^*, \text{st}^*)$, then $\mathcal{B}_1$ also terminates and outputs $(\text{com}^*, \text{st}^*)$. Finally, $\mathcal{B}_2$ behaves exactly the same as $\mathcal{A}_2$.

We also construct a PPT algorithm $\mathcal{D}$ against the $\text{hint-wHVZK}^\perp$ game. On input x, π with $\pi = (\pi_1, \dots, \pi_s)$, $\mathcal{D}$ runs $\mathcal{A}$ and responds the i -th query of $\mathbf{OTrans}^\perp$ by $\mathcal{A}_1$ with π_i . Finally, $\mathcal{D}$ outputs the same bit as the $\text{IMP-PA}^\perp$ game. By construction, we have

$$\begin{aligned} \text{Adv}_\Sigma^{\text{IMP-PA}^\perp}(\mathcal{A}) &= \Pr[1 \leftarrow \mathcal{D}(x, \pi) \mid (x, \pi) \leftarrow \text{Real}_\Sigma^\perp(1^\lambda, s)], \\ \text{Adv}_\Sigma^{\text{hint-IMP-PA}}(\mathcal{B}) &= \Pr[1 \leftarrow \mathcal{D}(x, \pi) \mid (x, \pi) \leftarrow \text{HintSim}_\Sigma^\perp(1^\lambda, s, \mathcal{S})]. \end{aligned}$$

Therefore, we have

$$\begin{aligned} \text{Adv}_{\Sigma, \mathcal{H}, \mathcal{S}}^{\text{hint-wHVZK}^\perp}(\mathcal{D}, s) &= \left| \text{Adv}_\Sigma^{\text{IMP-PA}^\perp}(\mathcal{A}) - \text{Adv}_\Sigma^{\text{hint-IMP-PA}}(\mathcal{B}) \right| \\ &\geq \text{Adv}_\Sigma^{\text{IMP-PA}^\perp}(\mathcal{A}) - \text{Adv}_\Sigma^{\text{hint-IMP-PA}}(\mathcal{B}). \end{aligned}$$

□

C Detailed Analysis on the Failure Probability

In this section, we analyze the failure probability of SQIsign2D-Push. The failure occurs when there exists no isogeny of degree $\in (0, 2^{e_1})_3$ between E_{com} and E_{chl} . Let $\mathcal{D}_{\text{com}}$ be the distribution of E_{com} computed by **FastCommit** and let $\mathcal{D}_{\text{chl}}(E_{\text{pk}})$ be the distribution of E_{chl} when $\text{pk} = E_{\text{pk}}$. Then we can define the failure probability as follows:

$$\mathcal{P}_{\text{pk}}^\perp = \Pr \left[\nexists \varphi : E_{\text{com}} \rightarrow E_{\text{chl}} \text{ of degree } \in (0, 2^{e_1})_3 \mid \begin{array}{l} E_{\text{com}} \leftarrow \mathcal{D}_{\text{com}}, \\ E_{\text{chl}} \leftarrow \mathcal{D}_{\text{chl}}(E_{\text{pk}}) \end{array} \right].$$

Note that this probability depends on pk . First, we perform a heuristic-based analysis, followed by a theoretical analysis. Finally, we show experimental results under the concrete parameter settings.

C.1 Failure Probability of the Response

Here, we analyze the failure probability using a heuristic. For supersingular elliptic curves E, E' over $\mathbb{F}_{p^2}$, we denote the smallest degree of isogeny $E \rightarrow E'$ by $d(E, E')$. From [13, Lemma 12], when $2^{e_1} \geq 2\sqrt{2p}/\pi$, it always satisfies that $d(E, E') < 2^{e_1}$. Therefore, we can bound the failure probability as follows:

$$\mathcal{P}_{\text{pk}}^\perp \leq \Pr[d(E_{\text{com}}, E_{\text{chl}}) \equiv 0 \pmod{3} \mid E_{\text{com}} \leftarrow \mathcal{D}_{\text{com}}, E_{\text{chl}} \leftarrow \mathcal{D}_{\text{chl}}(E_{\text{pk}})].$$

Now we assume the following heuristic.

Heuristic 2 *For any supersingular elliptic curve E' over $\mathbb{F}_{p^2}$, the distribution $\{d(E, E') \pmod{3} \mid E \leftarrow \mathcal{D}_{\text{com}}\}$ is uniform among $\{0, 1, 2\}$.*

Under this heuristic, we can bound $\mathcal{P}_{\text{pk}}^\perp$ by $1/3$ for any pk .

C.2 Theoretical Analysis

In this subsection, we analyze a theoretical upper bound on the failure probability. As in the main body of this paper, we identify elliptic curves that are isomorphic over $\mathbb{F}_p$, and let $\mathcal{E}$ be the set of supersingular elliptic curves over $\mathbb{F}_{p^2}$. To simplify the notation in this section, for a set S we write $|S|$ for its cardinality.

For ease of analysis, we assume that a commitment curve E_{com} and a challenge curve E_{chl} are uniformly and independently chosen from $\mathcal{E}$. For a real number $x > 0$, we consider the following probability:

$$\mathcal{P}_{< x} := \Pr[\exists \varphi : E \rightarrow E' \text{ s.t. } \deg \varphi < x, \deg \varphi \not\equiv 0 \pmod{3} \mid E, E' \leftarrow \mathcal{E}^{\text{unif}}],$$

where φ is an isogeny, $\mathcal{E}^{\text{unif}}$ is the uniform distribution over $\mathcal{E}$, and E, E' are independently chosen from $\mathcal{E}^{\text{unif}}$. We also define $\mathcal{P}_{< x}^\perp := 1 - \mathcal{P}_{< x}$. This is the probability that a response isogeny does not exist in our protocol assuming the uniformity of E_{com} and E_{chl} . Our goal is to give an upper bound on $\mathcal{P}_{< x}^\perp$. Our main result is the following theorem.

Theorem 5 *For a real number $x > 0$, we define a set $L_{x,3}$ as the set of primes ℓ such that $\ell < x$ and $\ell \nmid 3p$. If $\sum_{\ell \in L_{x,3}} (\ell + 1) \geq |\mathcal{E}|$, then we have*

$$\mathcal{P}_{<x}^\perp \leq \frac{2}{3}.$$

For our parameter settings, we use $p = c \cdot 2^{e_1} \cdot 3^{e_2} - 1$ and set $x = 2^{e_1}$. Therefore, we must check the condition $\sum_{\ell \in L_{x,3}} (\ell + 1) \geq |\mathcal{E}|$ for $x = 2^{e_1}$. We bound this sum as follows. By the result in [20, §4], we have $|L_{x,3}| \geq x/\ln x - 1$ for $x > 10$. Combining this with [36, Lemma 3.4], we obtain

$$\sum_{\ell \in L_{x,3}} (\ell + 1) > \frac{x^2}{\ln x} + \frac{x}{\ln x} - 4$$

for $x \geq 315437$. The right-hand side is easy to evaluate numerically. Thus, we can verify the condition using this lower bound. In particular, the parameters proposed in Section 6.1 satisfy it.

To prove Theorem 5, we prepare some lemmas. First, we give a lemma to count the number of collisions of a map between finite sets.

Lemma 3. *Let M, N be finite sets with $|M| > |N|$. For a map $f : M \rightarrow N$, we define a set*

$$C_f = \{(m_1, m_2) \in M \times M \mid m_1 \neq m_2, f(m_1) = f(m_2)\}.$$

Then, we have

$$|C_f| \geq \left(\frac{|M|}{|N|} - 1 \right) |M|$$

for any map $f : M \rightarrow N$.

Proof. Let $f : M \rightarrow N$ be a map. We define a set $C_f(n) = \{(m_1, m_2) \in M \times M \mid m_1 \neq m_2, f(m_1) = f(m_2) = n\}$ for $n \in N$. Then, we have $C_f = \bigsqcup_{n \in N} C_f(n)$ and $|C_f(n)| = |f^{-1}(n)|(|f^{-1}(n)| - 1)$. Therefore, we have

$$\begin{aligned} |C_f| &= \sum_{n \in N} |C_f(n)| = \sum_{n \in N} |f^{-1}(n)|(|f^{-1}(n)| - 1) \\ &= \sum_{n \in N} |f^{-1}(n)|^2 - \sum_{n \in N} |f^{-1}(n)| \\ &\geq \frac{1}{|N|} \left(\sum_{n \in N} |f^{-1}(n)| \right)^2 - |M| \\ &= \left(\frac{|M|}{|N|} - 1 \right) |M|. \end{aligned}$$

Here, the inequality follows from the convexity of the function $x \mapsto x^2$. □

Next, we give a lemma to count the number of endomorphisms of supersingular elliptic curves. For an integer $m > 0$ coprime with p , we define a set

$$\overline{\text{End}}(E; m) = \{\alpha \in \text{End}(E) \setminus \mathbb{Z} \mid \deg \alpha = m\} / \text{Aut}(E),$$

where the quotient means that we identify α and $\iota \circ \alpha$ for any $\iota \in \text{Aut}(E)$.

To bound $\sum_{E \in \mathcal{E}} |\overline{\text{End}}(E; m)|$, we recall the discussion in [24]. We choose an indexing of $\mathcal{E}$ as $\{E_1, \dots, E_n\}$ with $n = |\mathcal{E}|$. Then, we let $B(m)$ be the Brandt matrix of degree m with respect to this indexing, i.e., the (i, j) -entry of $B(m)$ is the number of isogenies of degree m from E_i to E_j up to post-composition with automorphisms of E_j . Then, we have

$$\sum_{E \in \mathcal{E}} |\overline{\text{End}}(E; m)| = \begin{cases} \text{Tr}(B(m)) - (n - \varepsilon_p) & \text{if } m \text{ is a square,} \\ \text{Tr}(B(m)) & \text{otherwise,} \end{cases}$$

where ε_p is the number of supersingular elliptic curves over $\mathbb{F}_{p^2}$ with extra automorphisms. The first case holds since the multiplication-by- $\sqrt{m}$ map has degree m and lies in $\overline{\text{End}}(E; m)$ if and only if E has extra automorphisms. From the discussion in [24], we have the following lemma.

Lemma 4. *Let ℓ_1, ℓ_2 be distinct primes different from p such that $\ell_1 < \ell_2$. Then, we have*

$$\begin{aligned} \text{Tr}(B(\ell_1^2)) - (n - \varepsilon_p) &\leq 2(\ell_1 + 1)^2, \\ \text{Tr}(B(\ell_1 \ell_2)) &\leq 2(\ell_1 + 1)(\ell_2 + 1). \end{aligned}$$

Proof. By the proof of Corollary 3.3 in [24], we have

$$\text{Tr}(B(\ell_1^2)) - n \leq 2\ell_1^2 + \frac{\ell_1}{2} + \frac{1}{6}.$$

The first inequality follows from this and $\varepsilon_p \leq 2$.

By the proof of Theorem 4.2 in [24], we have

$$\text{Tr}(B(\ell_1 \ell_2)) \leq 2\ell_2(\ell_1 + 1).$$

The second inequality follows directly from this.

Note that [24] assumes $p \equiv 1 \pmod{12}$, but the inequalities above hold for any prime p . $\square$

For a finite set L of primes different from p , we define a probability

$$\mathcal{P}(L) = \Pr[\exists \varphi : E \rightarrow E' \text{ s.t. } \deg \varphi \in L \mid E, E' \leftarrow \mathcal{E}^{\text{unif}}],$$

where φ is an isogeny, and E, E' are independently chosen from $\mathcal{E}^{\text{unif}}$. Then, we have the following lemma.

Lemma 5. *For a finite set L of primes different from p such that $\sum_{\ell \in L} (\ell + 1) \geq |\mathcal{E}|$, we have*

$$\mathcal{P}(L) \geq \frac{1}{3}.$$

Proof. For $E \in \mathcal{E}$, let $\overline{\text{Isog}}(E; L)$ be the set of isogenies from E such that $\deg \varphi \in L$ up to post-composition with isomorphisms. From each class in $\overline{\text{Isog}}(E; L)$, we choose one representative, and let $\text{Isog}(E; L)$ be the set of chosen representatives. Then, we have $|\text{Isog}(E; L)| = \sum_{\ell \in L} (\ell + 1)$. We denote by $\text{Codom}(E; L)$ the set of codomains of isogenies in $\text{Isog}(E; L)$.

Let f be the map from $\text{Isog}(E; L)$ to $\text{Codom}(E; L)$ that maps an isogeny to its codomain, and let C_f be the set defined in Lemma 3. We define a map

$$g : C_f \rightarrow \bigsqcup_{\substack{\ell_1, \ell_2 \in L \\ \ell_1 \leq \ell_2}} \overline{\text{End}}(E; \ell_1 \ell_2); \quad (\varphi_1, \varphi_2) \mapsto \widehat{\varphi_1} \circ \varphi_2.$$

Assume that two distinct pairs (φ_1, φ_2) and (φ'_1, φ'_2) in C_f has the same image under g . If $\deg \varphi_1 = \deg \varphi'_1$ then $\deg \varphi_2 = \deg \varphi'_2$. Let ℓ be $\deg \varphi_2$. Then we have $\ker \varphi_2 = \ker(\widehat{\varphi_1} \circ \varphi_2) \cap E[\ell]$ and the similar for φ'_2 . Therefore, we have $\ker \varphi_2 = \ker \varphi'_2$, thus, $\varphi_2 = \varphi'_2$. This implies $\varphi_1 = \varphi'_1$. If $\deg \varphi_1 \neq \deg \varphi'_1$ then $\deg \varphi_1 = \deg \varphi'_2$. Let ℓ be $\deg \varphi_1$. Then $\ker(\widehat{\varphi_1} \circ \varphi_2) \cap E[\ell]$ is equal to the kernel of the pull-back $[\varphi_2]^* \widehat{\varphi_1}$. Therefore, we have $[\varphi_2]^* \widehat{\varphi_1} = \varphi'_2$. A similar discussion shows that $[\varphi'_2]^* \widehat{\varphi'_1} = \varphi_2$. In summary, the inverse image $g^{-1}(g(\varphi_1, \varphi_2))$ is $\{(\varphi_1, \varphi_2)\}$ if $\deg \varphi_1 = \deg \varphi_2$, and $\{(\varphi_1, \varphi_2), ([\varphi_2]^* \widehat{\varphi_1})_* \varphi_2, [\varphi_2]^* \widehat{\varphi_1}\}$ otherwise. This implies that

$$|C_f| = \sum_{\ell \in L} |\overline{\text{End}}(E; \ell^2)| + 2 \sum_{\substack{\ell_1, \ell_2 \in L \\ \ell_1 < \ell_2}} |\overline{\text{End}}(E; \ell_1 \ell_2)|.$$

By Lemma 3, we have

$$\left(\frac{|\text{Isog}(E; L)|}{|\text{Codom}(E; L)|} - 1 \right) |\text{Isog}(E; L)| \leq |C_f|.$$

Combining these inequalities and summing over $E \in \mathcal{E}$, we have

$$\sum_{E \in \mathcal{E}} \left(\frac{|\text{Isog}(E; L)|}{|\text{Codom}(E; L)|} - 1 \right) |\text{Isog}(E; L)| \leq \sum_{\ell \in L} \sum_{E \in \mathcal{E}} |\overline{\text{End}}(E; \ell^2)| + 2 \sum_{\substack{\ell_1, \ell_2 \in L \\ \ell_1 < \ell_2}} \sum_{E \in \mathcal{E}} |\overline{\text{End}}(E; \ell_1 \ell_2)|.$$

By Lemma 4, the right-hand side is bounded above by

$$2 \sum_{\ell \in L} (\ell + 1)^2 + 4 \sum_{\substack{\ell_1, \ell_2 \in L \\ \ell_1 < \ell_2}} (\ell_1 + 1)(\ell_2 + 1) = 2 \left(\sum_{\ell \in L} (\ell + 1) \right)^2.$$

Since $|\text{Isog}(E; L)| = \sum_{\ell \in L} (\ell + 1)$ for any $E \in \mathcal{E}$, we have

$$\sum_{E \in \mathcal{E}} \frac{1}{|\text{Codom}(E; L)|} - \frac{|\mathcal{E}|}{\sum_{\ell \in L} (\ell + 1)} \leq 2.$$

By assumption, we have $|\mathcal{E}| / \sum_{\ell \in L} (\ell + 1) \leq 1$. Therefore, we have

$$\sum_{E \in \mathcal{E}} \frac{1}{|\text{Codom}(E; L)|} \leq 3.$$

Consequently, we have

$$\begin{aligned}\mathcal{P}(L) &= \frac{1}{|\mathcal{E}|^2} \sum_{E \in \mathcal{E}} |\text{Codom}(E; L)| \\ &\geq \frac{1}{|\mathcal{E}|} \left(\frac{1}{|\mathcal{E}|} \sum_{E \in \mathcal{E}} \frac{1}{|\text{Codom}(E; L)|} \right)^{-1} \\ &\geq \frac{1}{3}.\end{aligned}$$

Here, the first inequality follows from the concavity of the function $x \mapsto 1/x$. $\square$

Finally, we prove Theorem 5.

Proof (Proof of Theorem 5). Since $\mathcal{P}_{< x} \geq \mathcal{P}(L_{x,3})$, we have $\mathcal{P}_{< x}^\perp \leq 1 - \mathcal{P}(L_{x,3})$. By Lemma 5, we have $\mathcal{P}(L_{x,3}) \geq 1/3$. Therefore, we have $\mathcal{P}_{< x}^\perp \leq 2/3$. $\square$

Remark 4. The upper bound $2/3$ of the failure probability in Theorem 5 is far from tight. Indeed, our experiments in the next subsection suggest that the failure probability is at most approximately 2^{-18} for our parameter settings. One reason for this gap is that we only consider isogenies of prime degrees in Lemma 5. Allowing composite degrees would improve the upper bound, but the resulting analysis is considerably more involved. We leave tightening the bound to future work.

C.3 Experimental Results for the Failure Probability

Finally, we show in Table 5 our experimental results for the failure probability under the parameter settings show in Section 6.1. It has been experimentally confirmed that the failure probability is much smaller than we evaluate in the previous subsection. Note that no failures were recorded in the experiment for Level 5. Therefore, the failure probability is considered to be extremely small, but a more detailed evaluation is left for future work.

Table 5. Experimental failure count in 4,000,000 trials.

NIST level	e_1	e_2	c	# of failure	failure rate
1 ($\lambda = 128$)	131	78	1	1	$\approx 2^{-22}$
3 ($\lambda = 192$)	191	117	1	13	$\approx 2^{-18}$
5 ($\lambda = 256$)	263	156	1	0	0